

Dealing with Adversity: Religiosity or Science? Evidence from the Great Influenza Pandemic*

Enrico Berkes Davide M. Coluccia Gaia Dossi Mara P. Squicciarini

This Version: July, 2025

Abstract

How do societies respond to adversity? We provide novel evidence focusing on the reaction to the 1918–1919 influenza pandemic in the United States. We document that counties hit harder by the pandemic became both more religious and more scientific. We investigate the underlying mechanism, examining how individuals within counties reacted to the shock. We uncover heterogeneous responses, with religion and science acting as substitute ways of coping with the pandemic. Facing adversity widened the distance in religiosity between science-oriented individuals and the rest of the population and increased the polarization of religious beliefs.

JEL: J24, N13, Z12

Keywords: Religiosity, Science, Innovation, Influenza Pandemic

*We are grateful to Ran Abramitzky, Ainoa Aparicio Fenoll (discussant), Sascha Becker, Roland Bénabou, Davide Cantoni, Martin Fiszbein, Roberto Galbiati, Claudia Goldin, Reka Juhasz, Ben Marx, Joel Mokyr, Nathan Nunn, Thomas Piketty, Jared Rubin, Marco Tabellini, Felipe Valencia, John Van Reenen, Nico Voigtländer, Bruce Weinberg, Noam Yuchtman, and Katia Zhuravskaya for comments and discussions. We thank seminar audiences at the 2022 ASREC Annual Conference, the ASREC Workshop on the Economics of Religion, Barcelona, Bergen, Bicocca, Bocconi, Bolzano, Boston University, Brown, Geneva, Harvard, Louvain la Neuve, LSE, LUISS, Mannheim, the 6th Marco Fanno Alumni Workshop, Northwestern, Parthenope, Pavia, Pompeu Fabra, PSE/Sciences Po PEPES, Stanford, UC Berkeley, the University of British Columbia, Utrecht, and Warwick for feedback and suggestions. We also thank Luca Favero, Lorenzo Pedretti, Luisa Pomarici, and Sara Veronesi for excellent assistance during the construction of the dataset, and Ran Abramitzky, Andy Ferrara, and Marco Tabellini for kindly sharing data. Berkes: University of Maryland, Baltimore County, enrico.berkes@umbc.edu; Coluccia: University of Bristol, davide.coluccia@bristol.ac.uk; Dossi: Einaudi Institute for Economics and Finance, gaia.dossi@eief.it; Squicciarini: Bocconi University and CEPR, mara.squicciarini@unibocconi.it.

1 Introduction

Throughout history, the occurrence of adverse events—such as natural disasters and pandemics—has posed challenges to societies worldwide and continues to do so today. Understanding how individuals cope with adversity has key social, economic, and political implications and has been the focus of a vast literature in economics and other social sciences. Specifically, a strand of research documents that negative shocks lead to an increase in religiosity (Bentzen, 2019). Another strand finds that economies react by boosting scientific efforts (Miao and Popp, 2014; Moscona, 2022).¹

In this paper, we show that these two responses can occur *simultaneously*, making societies both more religious *and* more science-oriented—a finding at odds with the existing evidence documenting a negative relationship between religiosity and science (e.g., Bénabou, Ticchi, and Vindigni, 2015, 2022; Lecce, Ogliari, and Squicciarini, 2021). To investigate the mechanism behind this pattern, we study how individuals *within* society react to an adverse shock. We uncover heterogeneous responses, with religion and science acting as substitute ways through which different individuals react to adversity. These individual-level findings reconcile our aggregate results with the existing literature on the negative relationship between religiosity and science.

The setting of our study is the Great Influenza Pandemic (1918–1919) in the United States. Historical records document that many individuals strengthened their religious faith to cope with the pandemic. At the same time, the period following the pandemic saw increased scientific progress and fundamental medical advances.² To conduct our empirical analysis, we construct a novel data-driven measure of religiosity at a geographically disaggregated level. This measure is based on the naming patterns of babies born between 1900 and 1930 from the historical full-count censuses. Complementing this dataset with information from the Census of Religious Bodies, we empirically identify religious names and construct a measure of “revealed religiosity.” The underlying idea is that the first name given to a child conveys information on the religiosity of their parents. Our metrics of scientific progress are the share of people in science, technology, engineering, or mathematics (STEM) occupations and the universe of geo-coded patents granted in the U.S.³

Using a difference-in-differences framework, we first show that counties hit harder by the shock experienced an increase in religiosity. A one-standard-deviation increase in excess deaths—our main

¹For example, Bentzen (2019) documents that, across countries and within regions, individuals become more religious when hit by earthquakes. Moscona (2022) finds an increase in innovation efforts towards technologies that mitigate environmental distress in U.S. counties more exposed to the Dust Bowl during the 1930s.

²An increase in religiosity and scientific progress has also been documented after the COVID-19 outbreak. Bentzen (2021), using Google search data, finds a sharp increase in the intensity of prayers during the early days of the pandemic. Agarwal and Gaule (2022) show that the COVID-19 pandemic catalyzed R&D expenditure on pharmaceuticals and digital technologies.

³We refer to science and scientific progress interchangeably and use the two main proxies mentioned in the text.

measure for the intensity of the influenza shock—led to a 0.16 standard deviation increase in overall religiosity. We further document that these same counties also experienced an increase in scientific progress. We find that employment in scientific occupations grew in counties hit harder by the pandemic. Additionally, a one-standard-deviation increase in excess deaths led to a 0.17 standard deviation increase in overall patenting activity. Event-study analyses illustrate the absence of pretrends, providing further support for the validity of the research design. As a result of the contemporaneous increase of religiosity and science, their relationship turned from negative before the pandemic to positive afterwards. This is especially puzzling because it contrasts with the existing evidence documenting a negative relationship between the two (Bénabou et al., 2015, 2022; Lecce et al., 2021).

What is the mechanism behind the contemporaneous increase in religiosity and science? To answer this question, in the second part of the paper, we move to a within-county analysis and study individual-level reactions to the pandemic. We obtain three main results.

First, the shock increased the distance in religiosity between individuals in STEM and the rest of the population. Second, we document heterogeneous responses by religious background. In the aftermath of the pandemic, individuals from less religious backgrounds became more likely to select a scientific occupation. Those from more religious backgrounds became more religious.⁴ This pattern suggests that individuals coped with the negative shock in heterogeneous ways: some turned to science, while others turned to religion. These results also hold when restricting the sample to individuals in skilled occupations, suggesting that religious background is not merely proxying for the level of human capital or for the socio-economic background. Third, we document that the pandemic widened preexisting differences in religious sentiment: individuals from more (less) religious backgrounds became even more (less) religious. Consequently, in counties more exposed to the pandemic, the distribution of religiosity became more polarized.

The individual-level analysis sheds light on the drivers of the aggregate patterns and reconciles the county-level findings with the existing literature. While a county may have become more religious and innovative, individuals reacted differently to the same shock based on their scientific orientation or religious background. Science and religiosity appear to have been alternative ways of reacting to the pandemic. As a result, the distance in religiosity between science-oriented individuals and the others increased compared to before the shock, and the distribution of religious sentiment within society became more polarized.

We perform several checks to gauge the robustness of our findings. For both religiosity and scientific progress, we show that our results are robust to weighting regressions by county population and when dropping one state at a time from the analysis. They also hold when we run our analysis

⁴We measure religious background using individuals' own names (as opposed to their children's), aiming to capture the religious upbringing of a person instead of their current faith.

at the city level. These exercises suggest that our findings are not driven by small counties, specific states, or individuals residing in rural areas. Second, we perform a series of checks on our religiosity measure. In particular, we validate it internally across several dimensions (e.g., by computing our indicator excluding firstborn babies and accounting for potential heterogeneity in fertility patterns). In addition, we run a set of robustness checks to ensure that the increase in religiosity is not driven by migration.⁵ Next, we externally validate our data-driven measure of religiosity by using alternative indicators. In particular, we show that results are robust when using the share of biblical and saints' names and when using a non-name-based measure of religiosity, i.e., the share of people affiliated with a religious denomination. We also perform a series of robustness checks on our measure of innovative activity (e.g., we show that the increase in patenting was not driven by low-quality innovations). Finally, we address the concern that other factors may be related to the pandemic and may have contemporaneously affected the evolution of religiosity and science, confounding our results. To do so, we start by documenting that initial religiosity and scientific progress (as well as other demographic characteristics) are not related to the intensity of the shock. Then, using an event-study design, we show that religiosity and innovation were on a similar path across treated and control groups before the pandemic. Additionally, we rule out that a separate yet overlapping shock—World War I—may explain our findings.⁶ Taken together, our empirical results, supported by historical records, provide evidence that the influenza pandemic was conceivably the main driver behind the aggregate increases in both religiosity and scientific progress.

Concerning our within-county results, one key question is why some individuals became more religious while others selected a scientific occupation. Our findings on religiosity are in line with the religious coping hypothesis, which posits that religious faith can help deal with personal distress following a negative shock.⁷ What motivated people to turn to science is less obvious. We propose a broad interpretation of “scientific coping,” with individuals turning to science either to deal with their psychological distress—as in the case of religious coping—or to try to actively mitigate the negative (e.g., health- and economic-related) effects of the pandemic. An alternative explanation is that individuals turned to science because of increased labor demand in STEM occupations. While market forces

⁵In particular, we first run a placebo exercise where we test for the impact of the pandemic on the names of adults. The results show no impact of the shock on adults' names, which we interpret as evidence that the observed increase in religiosity was not driven by ex-ante more religious people moving to areas hit harder by the shock. Then, we show that our results hold when excluding from the estimation sample all those who, in the 1930 census, reside in a state different from the one where they were born.

⁶For a systematic overview of alternative mechanisms and the corresponding robustness, see the summary table in Appendix C.

⁷A complementary explanation could be that individuals turn to religion as an insurance mechanism against the negative economic effects of the pandemic. While we cannot fully exclude this channel, we believe it is unlikely (as discussed in Section 6).

may have played a role, the heterogeneity by religious background suggests that individuals' preexisting religiosity also shaped their decision to turn to science. Although our findings do not directly uncover the individual-level motivations behind these different behaviors, we document that people from different religious backgrounds reacted in different ways to the same shock. This increased the distance in religiosity between individuals in STEM and the rest of the population, and increased the polarization of religiosity within society.

Related Literature. This paper is closely related to the literature studying how societies react to negative shocks. Previous work has shown that, in accordance with the religious coping hypothesis (Pargament, 2001; Ano and Vasconcelles, 2005; Norenzayan and Hansen, 2006), natural disasters are associated with an increase in religiosity, both historically (e.g. Belloc, Drago, and Galbiati, 2016; Bentzen, 2019) and in contemporary scenarios (Sibley and Bulbulia, 2012; Bentzen, 2021).⁸ Another set of studies documents that economic crises (Babina, Bernstein, and Mezzanotti, 2023), wars (Gross and Sampat, 2021), climate change (Miao and Popp, 2014; Moscona, 2022; Clemens and Rogers, 2023), and pandemics (Gross and Sampat, 2021; Agarwal and Gaule, 2022) all shape innovation activity. To the best of our knowledge, this is the first study to provide evidence that natural disasters may foster a contemporaneous increase in religiosity *and* innovation, and also the first to document the ensuing polarization of religiosity within society.⁹

Additionally, we speak to the literature on the drivers of polarization within society. Most of the existing work in economics focuses on the causes of *political* polarization. A set of studies documents that economic shocks can lead to a rise in political polarization (e.g., Autor, Dorn, Hanson, and Majlesi, 2020). To the best of our knowledge, we are the first to document that a negative shock can lead to a polarization of religious beliefs.

This study also contributes to the broad literature on the economics of religion, pioneered by Weber (1905). In particular, we contribute to those studies that analyze the link between religiosity and science.¹⁰ While most papers adopt a historical (Deming, 2010; Mokyr, 2011), theoretical (Bénabou et al., 2022), or cross-sectional perspective (Bénabou et al., 2015, 2022), to our knowledge, we are the first to study the interaction between religion and science in a panel setting and to uncover the individual-level dynamics behind their coevolution.¹¹

⁸The religious coping hypothesis, first developed in the psychology literature, posits that people who are subject to economic and social shocks turn to religious faith as a coping device to deal with personal distress.

⁹Many studies have looked at the impact of natural disasters on, among others, social norms (e.g. Posch, 2022), migration (e.g. Boustan, Kahn, and Rhode, 2012), and economic activity (e.g. Boustan, Kahn, Rhode, and Yanguas, 2020), while other works have focused on how negative shocks may affect the direction of innovation (e.g. Hanlon, 2015).

¹⁰Other studies analyze the relationship between religion and accumulation of human capital, more broadly (Becker and Woessmann, 2009; Botticini and Eckstein, 2012; Squicciarini, 2020). For an overview of the literature on the economics of religion, see Iannaccone (1998), Iyer (2016), and Becker, Rubin, and Woessmann (2021).

¹¹Lecce et al. (2021) study how religiosity impacts the birth and migration of scientists in 19th-century French cantons,

Finally, we add to a growing literature that uses the informational content of names to measure individuals’ characteristics. Names have been used, for example, to measure race and ethnicity (Abramitzky, Boustán, and Eriksson, 2016; Fouka, 2019), individualism (Bazzi, Fiszbein, and Gebresilasse, 2020), socioeconomic background (Biavaschi, Giuliatti, and Siddique, 2017; Olivetti, Paserman, Salisbury, and Weber, 2020), nationalism (Jurajda and Kovač, 2021), and religiosity (Abramitzky et al., 2016; Andersen and Bentzen, 2022). While all of these papers assume a preexisting rule to classify names (e.g., whether one has a biblical or saint name or a name shared by a major religious figure), to the best of our knowledge, we are the first to identify the religiosity of names *directly from the data*.¹²

The paper is structured as follows. Section 2 describes the Great Influenza Pandemic in the United States and discusses the historical evidence on its effects on religiosity and innovation. In Section 3, we present the data and our indicator of religiosity. In Section 4, we explain the empirical strategy and the core results. In Section 5, we explore the possible mechanisms underlying our findings. Section 6 discusses the results and the limitations of the analysis. Section 7 concludes.

2 Historical Background

This section provides an overview of the Great Influenza Pandemic in the United States and how it influenced religion and scientific progress.

2.1 The Great Influenza Pandemic

Between 1918 and 1919, the Great Influenza Pandemic—also known as “Spanish Flu”¹³—led to the death of an estimated 40 million people worldwide (approximately 1 in 30 people). It was one of the deadliest natural disasters in modern times (Barro, Ursúa, and Weng, 2020). In the United States, the pandemic started in the spring of 1918 with sporadic outbreaks. Then, a second, more severe wave began in September 1918. The final wave started in January 1919, ending that spring. In total, it killed about 500,000 Americans, corresponding to 0.7% of the U.S. population (Crosby, 1989).¹⁴

Historical and modern accounts suggest that the pandemic hit across the U.S. quasi-randomly. Crosby (1989) writes that the states with the highest mortality displayed diverse geographical, climatic,

and Andersen and Bentzen (2022) show that individuals with religious names are less likely to become engineers, scientists, or doctors, and that cities with more religious individuals grew slower. However, none of these studies analyze how an adverse shock affects society’s heterogeneous response in terms of religion and science, and the underlying individual-level dynamics.

¹²For details on how we construct our religiosity measure, see Section 3.

¹³The Great Influenza Pandemic is popularly known as “Spanish Flu” because media in Spain—which was neutral during World War I (WWI)—were free to report news on this disease. Conversely, countries involved in WWI imposed press censorship on the topic. This gave the (incorrect) impression that Spain was either more severely hit by the disease or that the pandemic had originated in Spain.

¹⁴By comparison, COVID-19 caused 1.13 million deaths in the United States, approximately 0.3% of the U.S. population, between March 2020 and February 2023 (<https://covid.cdc.gov/covid-data-tracker/#datatracker-home>; accessed February 12, 2023).

and demographic characteristics. The pandemic hit with varying intensity within states as well. For example, in Minnesota, the death rate in Saint Paul was about 70% higher than in Minneapolis, despite the two cities being just 8 miles apart. In Ohio, Dayton experienced an 80% higher mortality rate than Columbus, even though the two cities had similar demographic characteristics (Huntington, 1923; Almond, 2006).

The infection was caused by strains of the A/H1N1 influenza virus, whose origin is still unknown. Neither antiviral drugs to treat the primary disease nor antibiotics to cure secondary bacterial infections were available. Doctors had to rely on an array of mostly ineffective—sometimes fatal—medicines such as aspirin and quinine (Spinney, 2018). It is debated whether nonpharmaceutical interventions (NPIs)—such as using masks, canceling public events, closing schools, and implementing isolation measures and quarantines—were effective in limiting the spread of the disease.¹⁵

2.2 The Pandemic and Religion

A large literature documents that individuals become more religious in response to adverse events. One explanation comes from the “religious coping hypothesis,” which posits that individuals turn to religious beliefs or practices as a way to cope with sudden dramatic circumstances (Pargament, 2001).¹⁶

The influenza pandemic inflicted substantial emotional and socioeconomic distress and could have acted as a powerful amplifier of religious sentiments (Phillips, 2020). Historical records document that spiritualism gained momentum in the aftermath of the pandemic. For instance, in the United States, modern evangelism benefited from the pandemic, as evidenced by a sharp rise in the circulation of evangelical magazines (Frost, 2020). Membership in Christian Science also soared during these years, reaching an all-time peak in the 1930s.¹⁷ Catholics and Orthodox Jews identified the influenza as a manifestation of divine anger, the expiation of which called for prayers.¹⁸

¹⁵Some authors assert that NPIs were effective in reducing mortality (e.g., Markel, Lipman, Navarro, Sloan, Michalsen, Stern, and Cetron, 2007; Berkes, Deschenes, Gaetani, Lin, and Severen, 2023), while others show that the effect of NPIs on overall deaths was small and statistically insignificant (e.g., Barro, 2022).

¹⁶For example, Bentzen (2019) documents that individuals become more religious when hit by earthquakes. Religion may also represent an insurance mechanism when negative shocks occur: Ager, Hansen, and Lønstrup (2016) shows that after the 1927 Great Mississippi Flood, demand for social insurance led to higher churchgoing, while Ager and Ciccone (2018) document that in the 19th-century United States, a larger share of the population was organized in religious communities in counties that were exposed to higher common agricultural risk.

¹⁷Christian Science, founded in 1879, is part of the religious movements belonging to the metaphysical family. It seeks to restore the healing and thaumaturgic virtues of primitive Christianity and has been associated with avoidance of mainstream medicine (Stark, 1998).

¹⁸On the other hand, a few groups of progressive Protestants called for a more scientific interpretation of the pandemic (Phillips, 2020).

2.3 The Pandemic and Science

Historical evidence suggests that the period after 1918 was one of sharp intellectual and scientific progress and that the Great Influenza Pandemic was particularly influential in shaping the development of medical sciences (Barry, 2020). Despite being ineffective during the pandemic, medicine evolved enormously in subsequent years. In 1928, Alexander Fleming discovered the medical use of penicillin in treating bacterial infections. By the 1930s, virology had become an established branch of medicine, and the first influenza vaccines were being developed (Spinney, 2018). During this time, medicine became more “scientific” and, hence, effective (Barry, 2020).

These advancements in medicine went hand in hand with increased trust in scientific progress. For instance, in her personal journal, Canadian author L. M. Montgomery wrote, “[...] the Spirit of God no longer works through the church for humanity. It did once but it has worn out its instrument and dropped it. Today it is working through Science” (Montgomery, 1924, p. 211). Barry (2020) argues that the pandemic was the key driver behind this paradigm shift because it fostered scientific thinking in the face of such a sudden and dramatic shock.

This overview suggests that the 1918–1919 pandemic fostered both scientific progress and religiosity—a result that might seem at odds with theoretical and empirical evidence, which depicts religion and science as opposing forces (e.g., Bénabou et al., 2015, 2022; Lecce et al., 2021). In this paper, we provide evidence that the influenza shock led to a simultaneous increase in religiosity and scientific progress at the aggregate level. We then reconcile this apparent puzzle by showing that it induced polarization within society, with some people turning to religion and others turning to science.

3 Data

To conduct our analysis, we construct a new dataset that combines information on religiosity, scientific progress, and the incidence of the Great Influenza Pandemic. This section describes the outcome variables and the main explanatory variables. Appendix A describes the data in detail. In the first part of the analysis, counties are the geographical unit of observation.¹⁹ In the second part of the analysis, we use individual-level data. Table B.1 provides descriptive statistics for the main variables.

3.1 Measuring Religiosity

The key challenge when studying religiosity is that it is not easy to measure it today as well as in the past. It is especially challenging to find a metric of religiosity that combines geographical granularity and high-frequency time variation.²⁰

¹⁹To address concerns related to counties changing their boundaries over time, we use 1920 counties as our geography of reference.

²⁰This lack of data is clear in historical settings—Squicciarini (2020), for instance, uses different measures of religiosity, available for only a limited set of points in time—but it poses substantial limitations to contemporary studies as well. Recent

In our analysis, we propose a novel measure of revealed religiosity based on the naming patterns of newborn babies. The motivating argument is that parents who give comparatively more religious names are more likely to be religious themselves. Therefore, naming patterns provide a measure of “revealed religiosity” of parents, rather than of the children themselves.²¹

We now describe how we compute the religiosity score associated with first names. The key advantage of this approach is that it allows us to obtain a disaggregated yearly measure of religiosity and to study its changes in the short-to-medium term. The metric we define is conceptually similar to that developed by Abramitzky et al. (2016) and Andersen and Bentzen (2022). They measure religiosity based, respectively, on whether children have biblical or saint names or names of a major religious figure. Our approach differs from theirs as we *empirically* identify religious names, using data on the entire population of newborns and existing indicators of religiosity.

3.1.1 *Estimating Religiosity Scores for First Names*

We use two main sources to compute religiosity scores. First, we construct naming patterns at the county-cohort level from the full-count U.S. censuses between 1900 and 1930 (Ruggles, Flood, Foster, Goeken, Pacas, Schouweiler, and Sobek, 2021). More precisely, we take the first name of all children born in the United States between 1896 and 1930 and collapse them at the name-county-cohort level, thus obtaining a panel of name-county pairs at a yearly frequency.²² Next, we use county-level data from the Census of Religious Bodies. This census—taken once every ten years between 1906 and 1936—allows us to construct, for every county and census decade, the share of people affiliated with any religious denomination.²³

To obtain the religiosity scores, we first compute the frequency of names. We denote with Name Frequency ^{k_{cd}} the number of children with name k born over the years $[d - 10, d)$ in county c . Then, we estimate the

papers use information from surveys such as the World Value Survey to measure religiosity (Bénabou et al., 2015, 2022). Yet, because waves are typically years apart, it is challenging to use survey-based measures to study the dynamics of religiosity at high time frequency.

²¹A corollary is that names carry information on the individual’s religious background: while we cannot infer that an individual called “Paul” is comparatively more religious than one named “Harold,” we assume that the parents of “Paul” are likely to be more religious than those of “Harold.”

²²A cohort is defined as all babies born in a given year. The first cohort in our sample comprises all babies born in 1896. The underlying rationale is that the first Census of Religious Bodies was published in 1906, and we consider the ten cohorts preceding that year.

²³To gather information on the number of religious members in each county, a report was obtained directly from local churches and congregations. The shares are computed as the number of individuals affiliated with these groups divided by each county’s population. Catholics and Protestants jointly account for more than 90% of the people enumerated by the census.

following equation:

$$y_{cd} = \alpha_c + \alpha_d + \sum_{k=1}^K \gamma^k \times \log(1 + \text{Name Frequency}_{cd}^k) + \varepsilon_{cd}, \quad (1)$$

where y denotes either the share of people affiliated to any denomination in county c in decade d ; d corresponds to the two pre-pandemic decades of the religious censuses (1906 and 1916); α_c and α_d are, respectively, county and decade fixed effects.²⁴ The term K is the total number of names that occur in at least 0.3% of the overall sample.²⁵ To measure name frequencies, we include all babies born within ten years before each pre-pandemic census. Hence, we restrict the sample to cohorts between 1896 and 1916. Then, we aggregate these frequencies from cohorts to decades to estimate equation (1).

We label the coefficient (γ^k) as the *religiosity score* associated with name k ; we interpret names with larger estimated religiosity scores ($\hat{\gamma}^k$) as conveying higher religiosity. Because equation (1) includes county fixed effects, larger religiosity scores are attached to names that become comparatively more frequent in counties that experienced larger increases in religiosity. In Figure 1, we report the estimated religiosity scores from equation (1), where the outcome variable is the share of people affiliated with any religious organization. The figure shows that typically religious-sounding names, such as “Joseph,” “Paul,” and “Elizabeth,” all feature positive and large estimated religiosity scores. Because our estimation method seeks to isolate *distinctively* religious names, relatively common ones such as “Mary” or “John” are associated with negative scores. In the case of “Mary,” for instance, its popularity during this period was such that religious and non-religious people alike used it, thus preventing it from being associated with distinctively religious people. Moreover, names with little connection to saints or biblical episodes are associated with negative religiosity scores. This is the case for Germanic names, such as “Bertha,” “George,” and “Harold,” and other non-religious names, such as “Pearl.”

3.1.2 A Yearly County-Level Measure of Religiosity

From equation (1), we obtain a set of estimated religiosity scores $\{\hat{\gamma}^k\}_{k=1}^K$, which we use to construct a *yearly* indicator of religiosity at the county level. More specifically, our synthetic measure of religiosity

²⁴In one of our robustness checks, we compute an alternative measure of religiosity that does not include any fixed effect. The results hold. Moreover, the results are robust to including the raw name frequencies in equation (1) or, alternatively, applying an inverse hyperbolic sine transformation and adding .01 to the frequency inside the log.

²⁵We follow Fouka (2020) and restrict the number of names included in equation (1) primarily to avoid overfitting. Fouka (2020) uses a threshold of 1,000 for a name to be included in the analysis. In our preferred specification, we consider all names whose share in our overall sample is at least 0.3% and run checks around this threshold to assess the robustness of our results. We include name frequencies in log to reduce their skewness and add one to avoid dropping all counties where there is at least one name with no newborn children (approximately 40% of the sample).

is defined as the predicted values of equation (1):

$$\hat{y}_{ct} = \sum_{k=1}^K \hat{\gamma}^k \times \log(1 + \text{Name Frequency}_{ct}^k), \quad (2)$$

where t denotes a cohort between 1900 and 1930. In addition, by considering religiosity scores associated with different denominations, we construct a synthetic series for Catholic and Protestant religiosity separately.

A concern about our religiosity indicator is how much variation in county religiosity names explain, net of that captured by fixed effects. In Appendix B, we provide several robustness and validation exercises for our synthetic measure. First, Figure B.1 provides county-binned scatters of synthetic and measured religiosity by denomination. The figure summarizes the results from two distinct exercises. Panel A shows in-sample correlations, comparing Census-measured and predicted religiosity in 1906 and 1916. Panel B compares synthetic and measured religiosity in 1926. We refer to this as an “out-of-sample” correlation, as data from the Censuses of Religious Bodies carried out after the pandemic are not used to estimate religiosity scores.²⁶ Both graphs show a positive correlation between actual and predicted religiosity. This exercise provides evidence that naming patterns capture meaningful variation in religiosity and helps validate our measure.

Next, following Abramitzky et al. (2016), we use biblical and saint names as an alternative name-based measure of religiosity. Finally, as an additional indicator of religiosity, we use the county-level share of the population with a religious affiliation (for all affiliations, and separately for Catholics and Protestants) recorded by the Census of Religious Bodies for 1906, 1916, and 1926.

3.2 Measuring Scientific Progress

We measure scientific progress at the local level using the share of individuals employed in STEM occupations. The rationale for this measure is that a STEM occupation requires that an individual has received a science-oriented education – which plausibly correlates with more favorable attitudes toward science and scientific progress at the local level (Bianchi and Giorcelli, 2020; Biasi, Deming, Moser, and Dillon, 2022). For each county and census year (1900 to 1930), we compute, respectively, the share of individuals employed in a medical occupation and the share of individuals employed in a STEM occupation. We compute these shares relative to (i) the entire population; (ii) the number of individuals employed in skilled occupations.²⁷

²⁶Our results do not change if we include data from the 1936 Census of Religious Bodies. However, growing discontent resulted in substantially lower reporting rates in the last Census for some religious groups. Following Stark (1992), we consider it less reliable and exclude it from our analysis.

²⁷The second measure increases the comparability of the control group with STEM individuals. Table B.2 lists the set of occupations that we label as medical (Panel A), as STEM (Panel B), and as skilled (Panel C). By construction,

We complement our measure of scientific progress by using patent data from the Comprehensive Universe of U.S. Patents (CUSP; Berkes (2018)). The CUSP contains information on the universe of U.S. patents issued between 1836 and 2015. The data for the time period considered in our paper (1900–1930) are extracted from digitized patent documents obtained from the U.S. Patent and Trademark Office. For the purpose of our analysis, we first assign each patent to a county, based on the residence of its inventor, and a year, based on the year in which the patent was filed. When a patent lists multiple inventors, we give equal weight to the location of each inventor. From the CUSP, we also collect the technology classes associated with each grant (according to the U.S. Patent Classification system) and assign them to technology groupings following the crosswalk provided by the National Bureau of Economic Research (Hall, Jaffe, and Trajtenberg, 2001).²⁸

3.3 Exposure to the Great Influenza Pandemic

To measure the incidence of the Great Influenza Pandemic across U.S. counties, we use mortality statistics assembled by the U.S. Department of Commerce. These were first collected in 1915 and, throughout the 1915–1918 period, they covered 40% of counties, accounting for more than 60% of the U.S. population. We follow the methodology developed by Beach, Clay, and Saavedra (2022) and measure mortality caused by the influenza pandemic as average yearly deaths during the pandemic (1918–1919) divided by average yearly deaths in the years before the pandemic (1915–1917). Formally, excess mortality in county c is defined as

$$\text{Excess Deaths}_c = \frac{\frac{1}{2} \sum_{t=1918}^{1919} \text{Deaths}_{ct}}{\frac{1}{3} \sum_{t'=1915}^{1917} \text{Deaths}_{ct'}}. \quad (3)$$

This measure represents our baseline treatment. We also report results from a categorical treatment variable equal to one if the baseline treatment (Excess Deaths_c) is above its median and zero otherwise. Figure 2 displays the geographical variation in the intensity of the pandemic in terms of excess deaths. We find that the severity of the pandemic varies substantially across counties, even geographically close ones. The rationale behind this measure is that—all else being equal—higher mortality during the pandemic relative to the period just before is likely due to the pandemic itself. A possible threat

medical occupations are also STEM occupations, and both are also skilled. The list of Medical and STEM occupations is manually constructed. The list of skilled occupations includes all broad occupational groups which include at least a STEM occupation. These groups of occupations are: “Professional, Technical;” “Managers, Officials, and Proprietors;” and “Craftsmen.” Compared to other occupations, skilled occupations are characterized by higher average returns. Their occupational income score—a variable that assigns to each occupation the median total income of all persons with that occupation in 1950—is, on average, two-thirds greater than that of other occupations.

²⁸We combine the “chemical” and “drugs” NBER classes into a single class, which we label “pharmaceuticals.” We follow this approach because of the large overlap between the number of chemical and drug patents at the county level, which would make it difficult to study them separately. All results for the pharmaceutical class also hold if we consider drug and chemical patents separately. An example of a pharmaceutical patent is shown in Figure B.2.

to this argument is the U.S. involvement in WWI and the fact that WWI deaths might confound our results. However, there is no significant correlation between WWI mortality and excess mortality, as shown in Figure B.3. In Section 4, we show that our results are robust to controlling for a post-1918 time indicator interacted with WWI-related deaths.

4 Main Results: County-Level Analysis

In this section, we present the baseline empirical strategy and show that exposure to the influenza pandemic led to an increase in both religiosity and scientific progress across counties.

4.1 Empirical Strategy

In the first part of the analysis, we separately study the impact of the pandemic on religiosity and scientific progress at the county level. Our sample consists of a panel of U.S. counties observed over the 1900–1929 period at decade or yearly frequency. In particular, we leverage quasi-random variation in exposure to the pandemic across U.S. counties in a difference-in-differences (DiD) setting and estimate the following specification:

$$y_{ct} = \alpha_c + \alpha_{s(c) \times t} + \mathbf{x}'_{ct} \boldsymbol{\beta} + \delta \times (\text{Post}_t \times \text{Excess Deaths}_c) + \varepsilon_{ct}, \quad (4)$$

where subscripts c and t denote county and time (decade or year), respectively; y_{ct} measures religiosity or scientific progress; α_c and $\alpha_{s(c) \times t}$ are county and state-by-time fixed effects; Post_t is an indicator variable equal to one if $t \geq 1918$ and zero otherwise; Excess Deaths_c measures the intensity of the pandemic in terms of excess deaths, as explained in Section 3.3; and ε_{ct} is the error term. In addition, in all regressions, we control for an interaction term between 1910 population and the post indicator ($\mathbf{x}'_{ct}$). Standard errors are clustered at the county level. Our coefficient of interest, δ , captures the impact of the pandemic on religiosity or scientific progress. To investigate possible heterogeneity of treatment effects over time, we also estimate a more flexible specification that, rather than interacting Excess Deaths with the Post indicator, interacts Excess Deaths with biennial time dummies:²⁹

$$y_{ct} = \alpha_c + \alpha_{s(c) \times t} + \mathbf{x}'_{ct} \boldsymbol{\beta} + \sum_{\substack{k=1909 \\ k \neq 1917}}^{1928} \delta^k \times I\{t = k \mid t = k + 1\} \times \text{Excess Deaths}_c + \varepsilon_{ct}, \quad (5)$$

where $I\{t = k \mid t = k + 1\}$ is an indicator variable that takes value one if t is in the two-year window indexed by k , and zero otherwise.

Did the influenza pandemic indeed spread randomly across counties? We perform three main

²⁹In the dynamic DiD specifications, we code periods over two-year windows to reduce noisy fluctuations in estimated treatment effects and to improve efficiency by pooling observations. We consider a 10-year period before and after 1917–18.

exercises to test this in the data. First, in Table B.3, we report the correlation between the intensity of the pandemic and religiosity, scientific progress, and a set of county covariates, all measured in 1910, the last census before the pandemic (or in the decade 1901–1910 in the case of yearly variables) accounting for population and state-level fixed effects.³⁰ Counties more exposed to the pandemic are observationally equivalent with respect to all variables except the share of foreigners. This aligns with the pandemic being comparatively more severe in urban areas, where immigrants were clustered.

Additionally, to rule out that these differences confound our analysis, we check whether control and treatment counties were on different trends before the shock by estimating event studies. Formally, in Equation (5), this implies that the estimates of δ^k would not be statistically different from zero before the pandemic hit, i.e., for all $k < 1917$.³¹ Our estimates support the parallel-trends assumption. However, this approach could still be invalid in the presence of shocks correlated with the intensity of the pandemic that positively affected both science and religiosity but that were *not* caused by the pandemic itself. A plausible candidate is the number of soldiers that counties lost in WWI: their deaths might have impacted the religiosity of their families or the ability (or motivation) of a county to produce innovation (or both). To test for this possibility, in Tables B.6 and B.13, we control for the number of deaths in WWI in our regression model and show that the results remain robust.

4.2 The Effect of the Influenza Pandemic on Religiosity

Table 1 shows the DiD estimates focusing on religiosity. In columns (1–3), the dependent variable is the share of individuals affiliated with any religious denomination (column 1), with a Catholic religious denomination (column 2), or with a Protestant religious denomination (column 3), from the Census of Religious Bodies. The estimates suggest that counties comparatively more exposed to the pandemic experienced an increase in religiosity, with no significant differences between Catholics and Protestants.

However, this measure of religiosity is available only at three points in time (1906, 1916, and 1926) and therefore does not allow us to study high-frequency variation in religiosity. In columns (4–6), we turn to our preferred measure of religiosity, the name-based index described in Section 3.1. This variable allows us to measure county-level religiosity every year between 1900 and 1929. We find that a one-standard-deviation increase in excess deaths led to a 0.16 standard deviation increase in name-based religiosity at the county level (column 4). Similarly, moving from a county at the 25th percentile

³⁰State fixed effects control for the fact that the pandemic spread from East to West between August 1918 and November 1918.

³¹Since the setting is not staggered—because the pandemic hit each county in the same period—equations (4) and (5) can be estimated through standard two-way fixed effects (Callaway and Sant’Anna, 2021; Sun and Abraham, 2021). Callaway, Goodman-Bacon, and Sant’Anna (2021), however, caution against using continuous treatments. We code a binary indicator equal to one for counties with above-median excess deaths. Throughout the paper, we show that the continuous and binary treatments yield qualitatively similar results.

of the excess mortality distribution to one at the 75th percentile led to an increase in religiosity of 10%.

In Figure 3, we report the coefficients of the interactions between the treatment variable and biennial dummies using the name-based measure of overall religiosity as dependent variable. The event study estimates support the patterns observed in the DiD analysis and confirm the absence of pre-trends. The increase in religiosity appears to persist over the decade after the pandemic. These findings are in line with the literature documenting a substantial persistence of religiosity (e.g., Squicciarini, 2020).

We perform a series of additional exercises to gauge the robustness of these findings. One potential concern is that our variation may be driven by small counties. Table B.4 replicates the specifications of Table 1, weighting regressions by county population in 1910. The baseline findings are confirmed.

Next, we run a set of robustness checks on our preferred specification, column (4) of Table 1. The results are shown in Table B.6. In column (1), we code the treatment as a binary variable equal to one if the baseline treatment is above its median and zero otherwise. In column (2), we control for mortality due to WW1. The estimated DiD coefficients remain positive and statistically significant throughout. In columns (3) and (4), we address the possibility that immigration confounds our results. In particular, one may be worried that comparatively more religious people moved into counties where the pandemic had been more severe, perhaps motivated by slacker labor markets. We address this concern in two ways. First, we exclude from the name-based religiosity measure all those individuals who, in the 1930 census, reside in a state different from where they were born. The results remain unchanged (column 3). Second, we compute a county-decade measure of religiosity based on the names of the adult population only. The in-migration mechanism would predict a positive impact of the pandemic on this variable. The estimates reported in column (4) show no evidence of any such effect, further ruling out that internal migration drives our results.

In a second set of checks, we test whether the results are robust to alternative ways of constructing the name-based religiosity measure. First, we show that our findings are robust to different ways of assigning name-based religiosity to households. One concern could be that firstborns are often named after a deceased grandparent. Thus, their names would reflect the higher religiosity of previous generations rather than their parents' religious attitudes. If, due to higher mortality, households in areas more affected by the pandemic were also more likely to have recently lost a grandparent, then our results might reflect a mechanical effect. Column (1) of Table B.6 reports estimates when dropping first-born children in every household. Another concern is that numerous households may display different naming behaviors for later-born children. In column (2), we drop children beyond the fourth. Then, if religious families displayed higher fertility rates, one may worry that our results are driven by an increase in the number of religious names due to the higher fertility of already religious households. In column (3), we compute within-household average religiosity to check whether our findings are

driven by larger households and differential fertility. All results hold through these alternative specifications. Next, in Table B.7 we explain the robustness of our measure to different estimating equations and frequency thresholds. In columns (1–2), we report the baseline result, but using religiosity scores estimated through equation (1) *without* county fixed effects. These scores are constructed based on the “stock” of religiosity in a given county instead of its deviations from the mean. The results are similar to our baseline estimates. Second, we test the robustness of our results to the number of names included in the sample. In our baseline analysis, we exclude names appearing in less than 0.3% of the overall population. Columns (3–8) in Table B.7 show that our findings are robust to choosing different frequency thresholds.

In a third set of checks, we perform our analysis using an alternative measure of religiosity. In Table B.8, we use biblical and saint names as an alternative name-based measure of religiosity, following Abramitzky et al. (2016). We find that the number of children named after biblical or saint names increased in areas more exposed to the pandemic after 1918. In addition, in Figure B.4, we show that the county-level share of Biblical and Saint names, computed using data from Abramitzky et al. (2016), is positively correlated with our baseline measure of religiosity.

In a fourth set of checks, we check whether specific states, or rural areas, might be driving our results. To rule out that our results are driven by specific states, we report treatment effects excluding from the sample one state at a time (Figure B.6a). The estimates remain remarkably stable and significant. To ensure that our results are not driven solely by rural areas, we study the effect of the pandemic at the city level. We use data on city-level deaths from Clay, Lewis, and Severnini (2019).³² The name-based religiosity measure is computed by leveraging variation in the naming patterns of children born in each city. In column (1) of Table B.9, we estimate equation (5) using a balanced panel of cities observed over the 1900–1929 period. Results are consistent with the baseline analysis: Religiosity increased in cities more severely affected by the pandemic. Moreover, the city-level sample includes several cities in Southern states, which are not included in our main analysis due to the lack of data on mortality.³³ We view the city-level analysis as shedding additional internal validity to the county-level analysis.

Finally, the validity of the research design requires that counties with different exposure to the pandemic were on parallel trends before 1918. While this assumption is hard to test, in Panel B.7a, we compare the confidence intervals of the event-study estimates with those obtained when allowing for per-period violations from linear trends up to a specific threshold (M), following Rambachan and Roth (2023). These show that the estimate remains significant at the 90% level for values of M below .1. This suggests that our results are robust to low power of pre-trends and nonlinear violations of

³²This dataset is discussed more in detail in Appendix A.8.

³³Figure B.5 reports the number of cities included in the sample by state and their location.

parallel trends.

Overall, we document that the pandemic led to an increase in religiosity through different specifications, samples, and measures of religiosity. This finding is consistent with the religious coping hypothesis, which posits that religion may serve as a coping device to deal with mental and psychological distress (e.g., Pargament, 2001; Bentzen, 2019, 2021). A complementary explanation could be that individuals turn to religion as an insurance mechanism against the negative effects of the pandemic. In Section 6, we discuss both explanations in more detail.

4.3 The Effect of the Influenza Pandemic on Scientific Progress

In this section, we study how the influenza pandemic impacted scientific progress. We show that this shock led to an increase in the share of people in STEM occupations and in patenting activity.

Table 2 (column 1) reports the results of estimating equation (4) on the share of individuals employed in STEM relative to the overall population. We perform the analysis at the decade level since this measure is taken from population censuses (1900, 1910, 1920, and 1930). We document an increase in the share of workers in STEM occupations in counties more severely hit by the pandemic. A one-standard-deviation increase in excess deaths is associated with a 0.985 standard deviation increase in the share of individuals in scientific occupations ten years later. Equivalently, moving from the 25th to the 75th percentile of the excess mortality distribution leads to a 31% increase in the share of individuals in STEM. In columns (2) and (3), we estimate the same specification as in column (1), but with, respectively, the share of individuals in medical occupations and the share of individuals in STEM non-medical occupations as the dependent variable. Similar results hold, suggesting that individuals reacted to the shock by turning to occupations close to medicine, but also to scientific occupations more broadly. Columns (4) to (6) replicate these specifications, focusing on the skilled sub-sample of the population, and provide similar results.

In Table 3, we study whether this increase in scientific occupation also translated into more innovation by focusing on county-level patent production. These allow us to construct a high-frequency indicator of scientific progress. Column (1) of Table 3 reports the results of estimating equation (4) on the total volume of innovation—measured as the $\log(1 + \text{number of patents})$ in a given county-year. We find that a one standard deviation increase in excess deaths led to a 0.17 standard deviation increase in the number of patents. Similarly, moving from a county at the 25th percentile to one at the 75th percentile of the excess-deaths distribution leads, on average, to an increase of 7% in the number of patents granted by county year. In column (2), we estimate the same specification on the $\log(1 + \text{number of patents in pharmaceuticals})$ and, in column (3), we estimate the same specification on the $\log(1 + \text{number of patents not in pharmaceuticals})$. The impact of the shock is positive in both cases, but it is larger in pharmaceuticals compared to the other fields (in standard deviations, it is twice the

magnitude). This suggests that the pandemic not only increased the volume of innovation but also induced a shift in its direction. In columns (4) to (6), we estimate the specifications of columns (1) to (3) on the sample of counties with at least one patent in every year of the sample. The magnitude is larger, and suggests that the increase in patenting comes both from the extensive margin (new counties starting to patent), but also from the intensive margin, i.e., an increase in patenting from counties that were already innovating before the pandemic.

Finally, Figure 4 shows the results shown in Table 3, column (5) in an event-study framework. Each dot in the plot reports the dynamic treatment effect of the pandemic on innovation in the indicated two-year window, as specified in equation (5). The coefficients suggest that the number of patents granted after the pandemic increased significantly more in more exposed counties, implying that the pandemic induced a sizable increase in innovative activities that persisted for at least one decade after the shock.³⁴

We run a series of robustness checks. First, as in the case of religiosity, one concern is that our results may have been driven by small counties where scientific production was relatively low during the pandemic. In Tables B.10 and B.11, we replicate the results of Tables 2 and 3, respectively, weighting each county by 1910 population. The results are similar to those of the main tables. Table B.12 deals with the potentially confounding role of immigration. In particular, we exclude from the estimation sample on occupational choice all those who, in the 1930 census, are recorded as residing in a state that is different from the one where they were born. The results remain unchanged.

Next, we focus on the patent data. In Table B.13, the dependent variable is the total number of patents. Column (1) reports results coding the treatment as a binary variable. Column (2) controls for WWI deaths interacted with the post-treatment indicator and confirms that WWI-related mortality is not driving our result. The estimated DiD coefficients remain positive and statistically significant throughout.

In the baseline specification, we take the logarithm of the number of patents and add one to avoid dropping zeros. In Table B.13, columns (3)–(6), we use alternative transformations of the dependent variable and obtain quantitatively similar findings. Additionally, we estimate the baseline equation as a Poisson Quasi-Maximum Likelihood regression. Similar results hold.

A plausible concern is that our results may be driven by “low-quality” innovation. Newspapers of the day often advocated remedies for influenza that were not science- or evidence-based, some of which may have been granted a patent. To address this concern, we use the text-based measures of “importance” developed by Kelly, Papanikolaou, Seru, and Taddy (2021).³⁵ Table B.14 shows the

³⁴The timing of the effect is in line with those described in Furman, Nagler, and Watzinger (2021) and Berkes and Nencka (Berkes and Nencka). These papers find that patent repository and Carnegie libraries, respectively, start having a positive and statistically significant impact on local patenting activity within the first year.

³⁵As discussed by Berkes (2018) and Andrews (2021), citation-based quality measures during this period are noisy and mostly uninformative due to the lack of a mandatory reference section until 1947. The measure built by Kelly et al. (2021)

results. In particular, we assign to every patent a dummy equal to one if the patent’s importance is in the top 20% of the distribution and zero otherwise. The number and share of these “breakthrough patents” substantially increase in counties hit harder by the pandemic.

As in the religiosity analysis, in Panel B.6b, we also report the results of estimating the impact of the pandemic on innovation excluding one state at a time. The coefficient remains stable, suggesting that our results are not driven by any particular state. Then, while most innovation activity clusters in urban areas, we perform our baseline analysis at the level of counties. To ensure that the results do not simply reflect rural-urban disparities, we estimate the effect of the pandemic on innovation at the city level. Column (2) in Table B.9 report the estimates of equation (1) for the panel of cities described in Section A.8. The results confirm the county-level evidence: despite the smaller sample size, we estimate the pandemic’s positive and statistically significant effect on the production of innovation.

Finally, in Panel B.7b, we confirm that the result remains remarkably robust to low power of pre-trends and non-linear violations of parallel trends.

Overall, we show that the pandemic led to an increase in scientific progress through different specifications, samples, and indicators. In line with the existing evidence (e.g., Moscona, 2022; Agarwal and Gaule, 2022), the shock also induced a change in the direction of occupational choice and of innovation. We document that, within the increase in STEM employment, individuals became disproportionately more likely to select medical occupations. For innovation, the pandemic led to an increase in overall as well as pharmaceutical innovation, plausibly in an attempt to mitigate the negative effects of the pandemic, or to work on preventing future health crises.

4.4 Joint Dynamics of Religiosity and Innovation

After studying the impact of the pandemic separately on religiosity and scientific progress, we now look at their joint evolution. Specifically, we test whether the *same* counties were affected along both dimensions or whether some counties saw an increase in religiosity while others saw an increase in scientific progress.

We estimate the following equation:

$$\begin{aligned}
y_{ct} = & \alpha_c + \alpha_{s(c) \times t} + \mathbf{x}'_{ct} \boldsymbol{\beta} + \delta_1 \times (\text{Excess Deaths}_c \times \text{Post}_t) + \delta_2 \times \text{Religiosity}_{ct} + \\
& + \delta_3 \times (\text{Religiosity}_{ct} \times \text{Post}_t) + \delta_4 \times (\text{Religiosity}_{ct} \times \text{Excess Deaths}_c) + \\
& + \delta_5 \times (\text{Excess Deaths}_c \times \text{Post}_t \times \text{Religiosity}_{ct}) + \varepsilon_{ct},
\end{aligned} \tag{6}$$

where y_{ct} is the number of patents divided by county population in 1910, following Bénabou et al. (2022), and Religiosity_{ct} is the religiosity measure described in Section 3.1. The coefficient δ_1 mea-

identifies important patents based on the textual similarity of a given patent to previous and subsequent work. Important patents are those that are distinct from previous work but are similar to subsequent innovations.

measures the impact of the pandemic on scientific progress, δ_2 captures the correlation between the outcome and religiosity before the pandemic, and the term δ_5 captures how the correlation between the outcome and religiosity changes after 1918 as a function of exposure to the pandemic. As before, the vector $\mathbf{x}_{ct}$ includes an interaction term between the county population in 1910 and a post-treatment indicator.

In Table B.15, we report the estimates of equation (6). The results suggest that counties comparatively more affected by the pandemic experienced a joint increase in religiosity and innovation. Columns (1) and (2) report the correlation between innovation and religiosity before and after the pandemic, respectively. Interestingly, this relationship shifts from negative to positive—as shown in Figure 5. Indeed, before the shock, there was a negative correlation between scientific progress and religiosity at the county level. This pattern aligns with contemporary evidence reported by Bénabou et al. (2015). After the pandemic, however, religiosity and science became positively correlated. This pattern is confirmed in column (3), which pools observations before and after the pandemic. In column (4), we estimate regression (6) and find that this shift in the correlation between religiosity and innovation co-moved with county-level exposure to the pandemic. In Section 5, we use individual-level data to uncover the possible mechanisms underlying these results.

5 Mechanisms: Individual-Level Analysis

After documenting that the pandemic led to a contemporaneous increase in religiosity and scientific production, two questions arise. Within counties, did the same—or different—individuals drive the increase in religiosity and scientific progress? And who in the population turned to religion, and who turned to science? In this section, we leverage individual-level data to answer these questions. In particular, we focus on individuals who were heads of household in the 1930 census.³⁶

First, we document that the pandemic widened the distance in religiosity between individuals in STEM occupations and the rest of the population. Second, we find that individuals from less religious backgrounds became more likely to select a STEM occupation, while those from more religious backgrounds increased their religiosity. As a result, the pandemic led to an increase in the polarization of religious beliefs within counties.

The within-county analysis reveals that the negative shock led to different reactions within society, with individuals becoming even more distant in terms of their scientific orientation and their pre-existing religiosity than they were before the pandemic. Additionally, it uncovers a negative relationship between religion and scientific progress at the individual level. This reconciles our aggregate findings with the existing literature and highlights important heterogeneity in individuals’ responses

³⁶The “head of household” variable is provided by the census. During this period, the father and/or husband was usually the head of the household whenever present.

to this negative shock.

5.1 The Pandemic Increased the Distance in Religiosity Between Science-Oriented Individuals and the Rest of the Population

We first test the impact of the influenza pandemic on the religiosity of science-oriented individuals. From full-count census data, for each individual, we use our measure of revealed religiosity based on the names of their children. Additionally, we leverage individual-level data on occupations.

We estimate the following triple-differences model:

$$\begin{aligned} \text{Religiosity}_{it} = & \alpha_{c(i) \times \text{STEM}(i)} + \alpha_{t \times \text{STEM}(i)} + \alpha_{c(i) \times t} + \mathbf{x}_i' \boldsymbol{\beta} + \\ & + \delta \times (\text{Excess Deaths}_{c(i)} \times \text{Post}_t \times \text{STEM}_i) + \varepsilon_{it}, \end{aligned} \quad (7)$$

The sample includes all children born between 1900 and 1929. Each observation (a child) is weighted by the inverse of the number of children in the household. The dependent variable is the religiosity associated with a child's name born in household i , living in county $c(i)$, in birth year t . All specifications include county-by-occupation ($\alpha_{c(i) \times \text{STEM}(i)}$), year-by-STEM occupation ($\alpha_{t \times \text{STEM}(i)}$), and county-by-year ($\alpha_{c(i) \times t}$) fixed effects. $\mathbf{x}_i$ is a vector of urban status, race of the household head, and birthplace of the household head. The coefficient δ compares STEM and non-STEM households, before and after the pandemic, by county-level exposure to the pandemic. $\text{Excess Deaths}_{c(i)}$ measures the intensity of the pandemic in terms of excess deaths, as explained in Section 3.3. Post_t is an indicator variable equal to one if the child was born in or after 1918, and zero otherwise. STEM_i is an indicator variable taking value one if the household head is employed in a STEM occupation in the 1930 census, and zero otherwise.

Table 4 shows the results. In column (1), we report results for the full population. In column (2), we restrict attention to heads of households employed in skilled occupations. In the aftermath of the pandemic, we find that individuals in STEM became less religious than non-STEM ones in counties more exposed to the shock. The effect is remarkably stable within the sample of individuals in skilled occupations, suggesting that the increased distance does not only reflect a broader socio-economic distance within society.

These results uncover that, while at the county level, we observe a contemporaneous increase in religiosity and scientific production, the pattern differs when we look at the reaction of individuals *within* counties. Specifically, in response to the pandemic, the distance in the religiosity of science-oriented individuals compared to that of the rest of the population increased, and this holds even in comparison to individuals employed in other skilled occupations. This suggests that, within the counties most affected by the influenza pandemic, *different* individuals drove the increase in religiosity and in scientific progress. Next, we examine *who* turned to science and *who* turned to religion in the

aftermath of the pandemic.

5.2 Turning to Science or Turning to Religion

In this section, we document that differences in individuals’ preexisting religious backgrounds shaped their response to the influenza pandemic. From full-count census data, we construct a measure of “background religiosity” for each individual based on their own name. Specifically, we interpret an individual’s own name as conveying information about the religiosity of their parents and, thus, the religious background of that individual.³⁷

We first show that an individual’s religious background shaped their propensity to work in a scientific occupation following the pandemic. To measure this, we use an indicator equal to one if an individual was employed in a STEM occupation and zero otherwise. Next, we document that an individual’s religious background also shaped their response to the pandemic in terms of religiosity.

We estimate two triple-difference specifications, one for the likelihood of selecting a STEM occupation and one for revealed religiosity.

To investigate the heterogeneous impact of the pandemic on occupational choice, we estimate the following equation:

$$\begin{aligned} \text{STEM}_i = & \alpha_{c(i) \times t} + \alpha_{c(i) \times B(i)} + \alpha_{B(i) \times t(i)} + \mathbf{x}_i' \boldsymbol{\beta} + \\ & + \delta_2 \times (\text{Excess Deaths}_{c(i)} \times \text{Post}_i \times \text{High Religious Background}_i) + \varepsilon_i, \end{aligned} \quad (8)$$

where i denotes an individual born in year $t(i)$. The unit of observation is an adult observed once in the 1930 census. The dependent variable is an indicator variable returning the value one if the head of household i is employed in a STEM occupation in 1930 and zero otherwise. $\text{Excess Deaths}_{c(i)}$ measures the intensity of the pandemic in terms of excess deaths, as explained in Section 3.3. Post_i is an indicator variable equal to one for individuals aged between 18 and 25 in 1918, and equal to zero otherwise. We define individuals as treated if they were 18 to 25 years old at the beginning of the pandemic for two reasons. First, according to the “impressionable years” hypothesis—a long-standing argument in psychology—economic, social, and cultural attitudes and beliefs are formed during early adulthood, approximately between the ages of 18 and 25, and change slowly after then (Giuliano and Spilimbergo, 2023). Second, the young cohort was plausibly more likely to adjust their employment choices as they were about to enter (or had recently entered) the labor market at the time. The background religiosity term $\text{High Religious Background}_i$ is an indicator variable taking value one if the religiosity score of the name of individual i is in the top 50% of the overall distribution and zero otherwise.

³⁷We restrict the analysis to U.S.-born individuals as we only assign background religiosity to individuals born in the United States.

To study the impact of the pandemic on religiosity, we estimate the following equation:

$$\begin{aligned} \text{Religiosity}_{it} = & \alpha_{c(i) \times t} + \alpha_{c(i) \times B(i)} + \alpha_{B(i) \times t} + \mathbf{x}'_i \boldsymbol{\beta} + \\ & + \delta_1 \times (\text{Excess Deaths}_{c(i)} \times \text{Post}_{it} \times \text{High Religious Background}_i) + \varepsilon_{it}, \end{aligned} \quad (9)$$

The sample includes children born between 1900 and 1929, and each observation (a child) is weighted by the inverse of the number of children in each household. The dependent variable Religiosity_{it} denotes the religiosity score of a child's name born in year t in household i , living in county $c(i)$. $\text{Excess Deaths}_{c(i)}$ measures the intensity of the pandemic in terms of excess deaths, as explained in Section 3.3. Post_i is a categorical variable equal to one for children born after 1918 and zero otherwise. High Religious Background is an indicator variable taking value one if the average religiosity of the head of household is equal to or above the median of background religiosity, or zero otherwise.

In equation 8 and equation 9, the terms $\alpha_{c(i) \times t}$, $\alpha_{c(i) \times B(i)}$, and $\alpha_{B(i) \times t}$ denote, respectively, county-by-year, county-by-high religious background, and year-by-high religious background fixed effects, and $\mathbf{x}_i$ includes urban status, race of the household head, and birthplace of the household head. The coefficients δ_1 and δ_2 quantify the effect of county-level exposure to the pandemic, comparing individuals in the top quintile of the background religiosity distribution with the rest of the population on, respectively, STEM employment and religiosity.

Table 5 presents the results of the analysis. In columns (1) and (3), the outcome variable is a dummy variable taking value one if the head of household is employed in a STEM occupation, or zero otherwise. In columns (2) and (4), the dependent variable is the individual's own religiosity. The main coefficient of interest is that on the interaction between the excess-deaths measure, the "Post" dummy, and the religious background of the household head. The findings suggest that an individual's religious background influenced their response to the pandemic. Specifically, those raised in less religious households became more likely to choose a scientific occupation in the face of adversity (column 1), while individuals raised by more religious parents became more likely to turn to religion (column 2). Notably, these patterns persist even when restricting the sample to individuals in skilled occupations (columns 3 and 4). This suggests that individuals' religious backgrounds shaped their response to the pandemic not only across socio-economic groups, but also within groups characterized by relatively high levels of human capital and economic resources. Further supporting this argument, we do not find evidence of a heterogeneous response by religious background in the likelihood of selecting a skilled occupation (Appendix Table B.16).

5.3 The Polarization of Religiosity

To conclude this section, we study the impact of the influenza pandemic on the *distribution* of religiosity. Specifically, we discretize the distribution of background religiosity into five quantiles, which

we label Q^{BR} .

We estimate the following equation:

$$\begin{aligned} \text{Religiosity}_{it} = & \alpha_{c(j) \times t} + \alpha_{c(j) \times Q(i)} + \alpha_{Q(i) \times t} + \mathbf{x}'_i \boldsymbol{\beta} + \\ & + \sum_{\substack{k=1 \\ k \neq 3}}^5 \delta^k \times [\text{Excess Deaths}_{c(i)} \times \text{Post}_{it} \times I(Q_i^{\text{BR}} = k)] + \varepsilon_{it}, \end{aligned} \quad (10)$$

The sample includes all children born between 1900 and 1929. Each observation (a child) is weighted by the inverse of the number of children in each household. The dependent variable measures the religiosity score associated with the name of a child born in household i living in county $c(i)$ in year t . The equation includes county-by-year ($\alpha_{c(j) \times t}$), county-by-quintiles of religious background ($\alpha_{c(j) \times Q(i)}$), and quintiles of religious background-by-year ($\alpha_{Q(i) \times t}$) fixed effects. The term $\mathbf{x}_i$ is a vector of urban status, race of the household head, and place of birth of the household head fixed effects. $\text{Excess Deaths}_{c(i)}$ measures the intensity of the pandemic in terms of excess deaths, as explained in Section 3.3. Post_{it} is an indicator variable equal to one if the child was born in or after 1918, and zero otherwise. For each quintile k , $I(Q_i^{\text{BR}} = k)$ is a dummy variable taking value one if the household background religiosity is in the k -th quintile of the distribution of background religiosity, equal to zero otherwise.

In Figure 6, we report the estimated $\{\delta^k\}$ coefficients. For each quintile of background religiosity k , the coefficient δ_k quantifies the impact of county-level exposure to the pandemic on the change in religiosity of individuals in quintile k relative to those in the third quintile (the baseline).³⁸

The results provide evidence of an increase in polarization: the higher the level of initial religious backgrounds, the larger the increase in religiosity in response to the pandemic. This pattern suggests that, within the same county, individuals from different religious backgrounds became even more distant in terms of their religiosity, increasing the polarization of religious sentiment within society.

Our findings reveal that individuals from more religious backgrounds, who were already more religious before the pandemic, became even more religious afterwards in more exposed counties.³⁹

The individual-level results help us uncover the channels behind the contemporaneous increase in religiosity and science at the county level. They suggest that, within counties, individuals reacted differently to the same shock, based on their pre-pandemic scientific orientation or their religious background. As a result, while a county may have become more religious *and* more innovative, individuals seemed to turn either to religion *or* to science, leading to within-county polarization of religiosity.

³⁸The associated regression estimates are reported in Appendix Table B.17.

³⁹The correlation between revealed religiosity and background religiosity is equal to 0.13 and highly statistically significant ($p < .001$), in line with a large literature on cultural transmission (Bisin and Verdier, 2001).

6 Discussion: Interpretation and Limitations of the Results

Our analysis shows two clear patterns: (i) the 1918–1919 influenza pandemic led to an increase in religiosity and scientific progress across U.S. counties and, as a result of the shock, the same counties became both more religious and more scientific; (ii) *within* counties, there was a heterogeneous response to the same shock, with some individuals turning to religion and others turning to science.

One concern is that other factors related to the pandemic may have affected the evolution of religiosity and science, confounding our results. To address this concern, we proceed in three steps. First, we document that neither initial religiosity, scientific progress, nor other demographic factors, are related to the intensity of the shock. Second, our event-study analysis shows the absence of pretrends, suggesting that religiosity and science were on a similar path in treated and control groups before the pandemic. Third, we account for other potentially confounding characteristics, such as differential fertility, WWI deaths, and migration patterns. Our results are robust in all these cases. The empirical evidence, supported by historical records, makes it hard to imagine that the pandemic did not trigger an increase in religiosity and scientific progress.

A second concern regards our main measures of religiosity and scientific progress. First, does our name-based indicator indeed capture religiosity at the local level? We show that our results are robust to alternative ways of constructing the name-based measure and to alternative classifications of religious names. In addition, in counties hit harder by influenza, the share of people affiliated with a religious denomination increased, providing further evidence that the pandemic led to an increase in local religiosity. Similarly, we use two measures for scientific progress: the share of individuals in scientific occupations and the total number of patents. While each measure has its caveats, they provide consistent and robust results.

One puzzle emerging when looking at the aggregate patterns is whether these results are driven by the same individuals becoming more religious and innovative or by different individuals reacting differently to the same shock. Our findings suggest that the second mechanism is at play. We document that the shock widened the distance in religiosity between science-oriented individuals and the rest of the population. In addition, we show that individuals from less religious backgrounds became more likely to choose a scientific occupation, while those from more religious backgrounds became even more religious. This suggests that a group of individuals turned to science, while a separate group turned to religion. As a result, the pandemic increased the polarization of religiosity within society: individuals from more (less) religious backgrounds became even more (less) religious.

One key question regarding our individual-level results is what explains the increase in religiosity or the choice of a scientific occupation. The findings on religiosity are consistent with the religious coping hypothesis, which suggests that religious faith can represent a coping device to deal with per-

sonal distress following a negative shock. A complementary explanation for why individuals may turn to religion is social insurance. While we cannot fully distinguish between the two (and it goes beyond the scope of our paper), we read our evidence as favoring the religious coping hypothesis. First, this is in line with the literature showing that intrinsic religiosity responds to unexpected events, as noted by Bentzen (2019). Second, as the increase in religiosity persists up to ten years after the shock, it is more likely to be related to a change in beliefs than to a temporary increase in the need for social insurance.

What motivates people to turn to science is less obvious. Individuals may turn to science to deal with their psychological distress, similarly to religious coping, or in an attempt to actively mitigate the negative (health-related or economic) effects of the pandemic. Another possibility is that individuals turn to science because of increased labor demand in STEM occupations. While we do not exclude that market forces mattered in this setting, our results suggest that individuals' religious backgrounds also played a key role in the decision to turn to science. These results also hold when restricting the sample to individuals in skilled occupations, suggesting that religious background is not merely proxying for the level of human capital or for the socio-economic background. While our findings cannot directly speak to the individual-level motivations behind these different behaviors, they provide evidence of a heterogeneous response to the same adverse event and of a polarization of religious attitudes within society.

Taken together, we interpret our results as evidence that individuals from religious backgrounds turned to religion as a coping device in the aftermath of the pandemic, while those from a scientific background turned to science.

7 Conclusions

In this paper, we provide new evidence on how societies respond to adversity, studying an exemplary historical episode: the Great Influenza Pandemic of 1918–1919.

First, we show that society reacted to the pandemic by becoming more religious as well as more scientific. Second, using individual-level data from full-count censuses, we find that the shock increased the distance in religiosity between science-oriented individuals and the rest of the population. Third, we show that individuals from less religious backgrounds became more likely to choose STEM occupations in response to the pandemic, while those from initially more religious backgrounds became more religious. As a consequence, the distribution of religiosity in counties more exposed to the pandemic became more polarized. These results suggest that religiosity and science acted as substitute ways of reacting to the shock.

Our paper sheds new light on the relationship between religiosity and science. Throughout history, science and religion have often been in conflict, and recent evidence by Bénabou et al. (2015, 2022)

shows that the two are negatively correlated, both across countries and across U.S. states. We provide novel evidence that—at the individual level—the two are substitute ways of dealing with adversity.

Our findings provide insights that can help explain modern events such as the reaction of society to the COVID-19 pandemic. Even though the modern context differs in many ways from the period of the influenza pandemic, including the medical advancements of the past century, the reaction of today's society has many similarities with what we document for the 1918–1919 pandemic.⁴⁰ In particular, our findings can help explain the opposing views that have emerged since the COVID-19 pandemic on science-based responses to the shock, such as the opposing attitudes toward vaccines.

Finally, our results suggest that, in the aftermath of a negative shock, societies may become more divided, providing novel evidence on the drivers of polarization within societies.

References

- Abramitzky, R., L. P. Boustan, and K. Eriksson (2016). Cultural Assimilation During the Age of Mass Migration. *NBER Working Paper* (No. w22381).
- Agarwal, R. and P. Gaule (2022). What Drives Innovation? Lessons from COVID-19 R&D. *Journal of Health Economics* 82, 102591.
- Ager, P. and A. Ciccone (2018). Agricultural Risk and the Spread of Religious Communities. *Journal of the European Economic Association* 16(4), 1021–1068.
- Ager, P., C. W. Hansen, and L. Lønstrup (2016). Church Membership and Social Insurance: Evidence from the Great Mississippi Flood of 1927. *Southern Denmark University Discussion Papers on Business and Economics*.
- Almond, D. (2006). Is the 1918 Influenza Pandemic Over? Long-term Effects of in Utero Influenza Exposure in the Post-1940 US Population. *Journal of Political Economy* 114(4), 672–712.
- Andersen, L. H. and J. S. Bentzen (2022). In the Name of God! Religiosity and the Transition to Modern Economic Growth. *CEPR Discussion Paper* (DP16938).
- Andrews, M. (2021). Historical Patent Data: A Practitioner's Guide. *Journal of Economics and Management Strategy* 30(2), 368–397.
- Ano, G. G. and E. B. Vasconcelles (2005). Religious Coping and Psychological Adjustment to Stress: A Meta-Analysis. *Journal of Clinical Psychology* 61(4), 461–480.
- Autor, D., D. Dorn, G. Hanson, and K. Majlesi (2020). Importing Political Polarization? The Electoral Consequences of Rising Trade Exposure. *American Economic Review* 110(10), 3139–3183.
- Babina, T., A. Bernstein, and F. Mezzanotti (2023). Financial Disruptions and the Organization of Innovation: Evidence from the Great Depression. *The Review of Financial Studies* 36(11), 4271–

⁴⁰One key difference between the two pandemics is that no medical remedy or vaccine became available until many years after the earlier pandemic ended.

4317.

- Barro, R. J. (2022). Non-pharmaceutical Interventions and Mortality in US Cities During the Great Influenza Pandemic, 1918–1919. *Research in Economics* 76(2), 93–106.
- Barro, R. J., J. F. Ursúa, and J. Weng (2020). The Coronavirus and the Great Influenza Pandemic: Lessons from the “Spanish Flu” for the Coronavirus’s Potential Effects on Mortality and Economic Activity. *NBER Working Paper* (No. w26866).
- Barry, J. M. (2020). *The Great Influenza: The Story of the Deadliest Pandemic in History*. London (UK): Penguin.
- Bazzi, S., M. Fiszbein, and M. Gebresilas (2020). Frontier Culture: The Roots and Persistence of “Rugged Individualism” in the United States. *Econometrica* 88(6), 2329–2368.
- Beach, B., K. Clay, and M. Saavedra (2022). The 1918 Influenza Pandemic and Its Lessons for COVID-19. *Journal of Economic Literature* 60(1), 41–84.
- Becker, S. O., J. Rubin, and L. Woessmann (2021). Religion in Economic History: A Survey. *The Handbook of Historical Economics*, 585–639.
- Becker, S. O. and L. Woessmann (2009). Was Weber Wrong? A Human Capital Theory of Protestant Economic History. *The Quarterly Journal of Economics* 124(2), 531–596.
- Belloc, M., F. Drago, and R. Galbiati (2016). Earthquakes, Religion, and Transition to Self-Government in Italian Cities. *The Quarterly Journal of Economics* 131(4), 1875–1926.
- Bénabou, R., D. Ticchi, and A. Vindigni (2015). Religion and Innovation. *American Economic Review: Papers & Proceedings* 105(5), 346–351.
- Bénabou, R., D. Ticchi, and A. Vindigni (2022). Forbidden Fruits: The Political Economy of Science, Religion and Growth. *The Review of Economic Studies* 89(4), 1785–1832.
- Bentzen, J. (2019). Acts of God? Religiosity and Natural Disasters Across Subnational World Districts. *The Economic Journal* 126(622), 2295–2321.
- Bentzen, J. S. (2021). In Crisis, We Pray: Religiosity and the COVID-19 Pandemic. *Journal of Economic Behavior & Organization* 192, 541–583.
- Berkes, E. (2018). Comprehensive Universe of US Patents (CUSP): Data and Facts. *Working Paper*.
- Berkes, E., O. Deschenes, R. Gaetani, J. Lin, and C. Severen (2023). Lockdowns and Innovation: Evidence from the 1918 Flu Pandemic. *Review of Economics and Statistics*, 1–30.
- Berkes, E. and P. Nencka. Knowledge access: The effects of Carnegie libraries on innovation. *The Review of Economics and Statistics*.
- Bianchi, N. and M. Giorelli (2020). Scientific Education and Innovation: from Technical Diplomas to University STEM Degrees. *Journal of the European Economic Association* 18(5), 2608–2646.
- Biasi, B., D. Deming, P. Moser, and E. W. Dillon (2022). *Education and Innovation*, pp. 537–558. Chicago: University of Chicago Press.

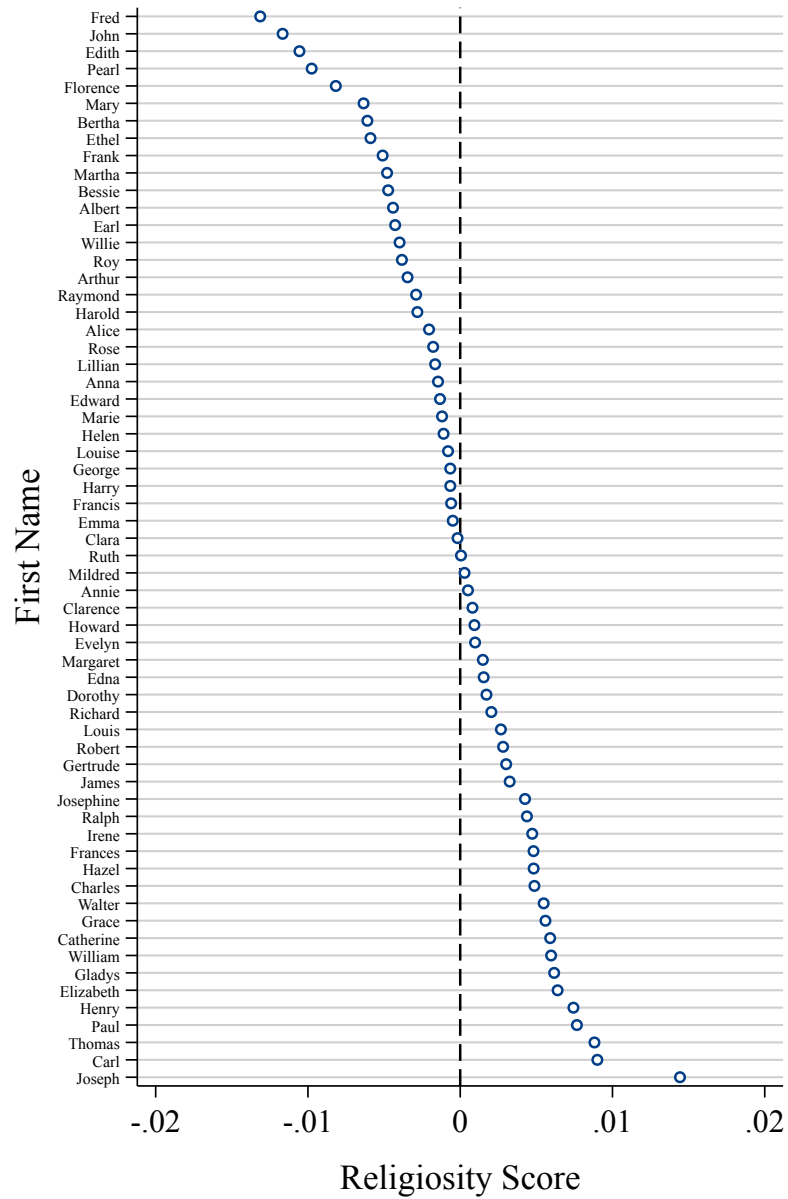
- Biavaschi, C., C. Giulietti, and Z. Siddique (2017). The Economic Payoff of Name Americanization. *Journal of Labor Economics* 35(4), 1089–1116.
- Bisin, A. and T. Verdier (2001). The Economics of Cultural Transmission and the Dynamics of Preferences. *Journal of Economic Theory* 97(2), 298–319.
- Botticini, M. and Z. Eckstein (2012). *The Chosen Few: How Education Shaped Jewish History, 70-1492*. Princeton University Press.
- Boustan, L. P., M. E. Kahn, and P. W. Rhode (2012). Moving to Higher Ground: Migration Response to Natural Disasters in the Early Twentieth Century. *American Economic Review* 102(3), 238–244.
- Boustan, L. P., M. E. Kahn, P. W. Rhode, and M. L. Yanguas (2020). The Effect of Natural Disasters on Economic Activity in US Counties: A Century of Data. *Journal of Urban Economics* 118, 103257.
- Callaway, B., A. Goodman-Bacon, and P. H. Sant’Anna (2021). Difference-in-Differences with a Continuous Treatment. *Working Paper*.
- Callaway, B. and P. H. Sant’Anna (2021). Difference-in-Differences with Multiple Time Periods. *Journal of Econometrics* 225(2), 200–230.
- Clay, K., J. Lewis, and E. Severnini (2019). What Explains Cross-City Variation in Mortality During the 1918 Influenza Pandemic? Evidence from 438 US Cities. *Economics & Human Biology* 35, 42–50.
- Clemens, J. and P. Rogers (2023). Demand Shocks, Procurement Policies, and the Nature of Medical Innovation: Evidence from Wartime Prosthetic Device Patents. *Review of Economics and Statistics*, 1–45.
- Crosby, A. W. (1989). *America’s Forgotten Pandemic: The Influenza of 1918*. Cambridge: Cambridge University Press.
- Deming, D. (2010). *Science and Technology in World History: Early Christianity, the Rise of Islam and the Middle Ages: 2*. Jefferson, NC: McFarland.
- Eckert, F., A. Gvartz, J. Liang, and M. Peters (2018). A Consistent County-Level Crosswalk for US Spatial Data since 1790. *NBER Working Paper* (No. w26632).
- Ferrara, A. and P. Fishback (2024). Discrimination, Migration, and Economic Outcomes: Evidence from World War I. *Review of Economics and Statistics* 106(5), 1201–1219.
- Fouka, V. (2019). How do Immigrants Respond to Discrimination? The Case of Germans in the US during World War I. *American Political Science Review* 113(2), 405–422.
- Fouka, V. (2020). Backlash: The Unintended Effects of Language Prohibition in US Schools after World War I. *The Review of Economic Studies* 87(1), 204–239.
- Frost, P. (2020). An Accelerant of Social Change? The Spanish Flu of 1918-19. *International Political Anthropology Journal* 13(2), 123–133.
- Furman, J. L., M. Nagler, and M. Watzinger (2021, November). Disclosure and Subsequent Innovation:

- Evidence from the Patent Depository Library Program. *American Economic Journal: Economic Policy* 13(4), 239–70.
- Giuliano, P. and A. Spilimbergo (2023). Recessions, Lifetime Experiences and the Formation of Political Beliefs. *Working paper*.
- Gross, D. P. and B. N. Sampat (2021). The Economics of Crisis Innovation Policy: A Historical Perspective. *American Economic Association: Papers and Proceedings* 111, 346–50.
- Hall, B. H., A. B. Jaffe, and M. Trajtenberg (2001). The NBER Patent Citation Data File: Lessons, Insights and Methodological Tools.
- Hanlon, W. W. (2015). Necessity is the Mother of Invention: Input Supplies and Directed Technical Change. *Econometrica* 83(1), 67–100.
- Huntington, E. (1923). Causes of Geographical Variation in the Influenza Epidemic in the Cities of the United States. *Bull. Nat. Res. Council* 6, 1–36.
- Iannaccone, L. (1998). Introduction to the Economics of Religion. *Journal of Economic Literature* 36(3), 1465–1496.
- Iyer, S. (2016). The New Economics of Religion. *Journal of Economic Literature* 54(2), 395–441.
- Jurajda, Š. and D. Kovač (2021). Names and Behavior in a War. *Journal of Population Economics* 34(1), 1–33.
- Kelly, B., D. Papanikolaou, A. Seru, and M. Taddy (2021). Measuring Technological Innovation Over the Long Run. *American Economic Review: Insights* 3(3), 303–20.
- Lecce, G., L. Ogliari, and M. P. Squicciarini (2021). Birth and Migration of Scientists: Does Religiosity Matter? Evidence from 19th-Century France. *Journal of Economic Behavior & Organization* 187, 274–289.
- Long, J. and J. Ferrie (2013). Intergenerational Occupational Mobility in Great Britain and the United States since 1850. *American Economic Review* 103(4), 1109–1137.
- Markel, H., H. B. Lipman, J. A. Navarro, A. Sloan, J. R. Michalsen, A. M. Stern, and M. S. Cetron (2007). Nonpharmaceutical Interventions Implemented by US Cities During the 1918-1919 Influenza Pandemic. *Journal of the American Medical Association* 298(6), 644–654.
- Miao, Q. and D. Popp (2014). Necessity as the Mother of Invention: Innovative Responses to Natural Disasters. *Journal of Environmental Economics and Management* 68(2), 280–295.
- Mokyr, J. (2011). The Economics of Being Jewish. *Critical Review* 23(1-2), 195–206.
- Montgomery, L. M. (1992[1924]). In M. Rubio and E. Waterston (Eds.), *The Selected Journals of L.M. Montgomery, Volume III: 1921-1929*. Toronto: Oxford University Press.
- Moscona, J. (2022). Environmental Catastrophe and the Direction of Invention: Evidence from the American Dust Bowl. *Working Paper*.
- Norenzayan, A. and I. G. Hansen (2006). Belief in Supernatural Agents in the Face of Death. *Person-*

- ality and Social Psychology Bulletin* 32(2), 174–187.
- Olivetti, C., M. D. Paserman, L. Salisbury, and E. A. Weber (2020). Who Married, (to) Whom, and Where? Trends in Marriage in the United States, 1850–1940. *NBER Working Paper* 28033.
- Pargament, K. I. (2001). *The Psychology of Religion and Coping: Theory, Research, Practice*. New York: Guilford Press.
- Phillips, H. (2020). ‘17, ‘18, ‘19: Religion and Science in three Pandemics, 1817, 1918, and 2019. *Journal of Global History* 15(3), 434–443.
- Posch, M. (2022). Do Disasters Affect the Tightness of Social Norms?
- Rambachan, A. and J. Roth (2023). A More Credible Approach to Parallel Trends. *Review of Economic Studies* 90(5), 2555–2591.
- Ruggles, S., S. Flood, S. Foster, R. Goeken, J. Pacas, M. Schouweiler, and M. Sobek (2021). IPUMS USA: Version 11.0 [dataset]. [Machine-readable database].
- Sibley, C. G. and J. Bulbulia (2012). Faith After an Earthquake: A Longitudinal Study of Religion and Perceived Health Before and After the 2011 Christchurch New Zealand Earthquake. *PloS One* 7(12).
- Spinney, L. (2018). *Pale Rider: the Spanish Flu of 1918 and How It Changed the World*. London: Vintage.
- Squicciarini, M. P. (2020). Devotion and Development: Religiosity, Education, and Economic Progress in 19th-Century France. *American Economic Review* (110), 3454–3491.
- Stark, R. (1992). The Reliability of Historical United States Census Data on Religion. *Sociological Analysis* 53(1), 91–95.
- Stark, R. (1998). The Rise and Fall of Christian Science. *Journal of Contemporary Religion* 13(2), 189–214.
- Sun, L. and S. Abraham (2021). Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects. *Journal of Econometrics* 225(2), 175–199.
- Weber, M. (1905). *The Protestant Ethic and the Spirit of Capitalism*. London: Routledge.

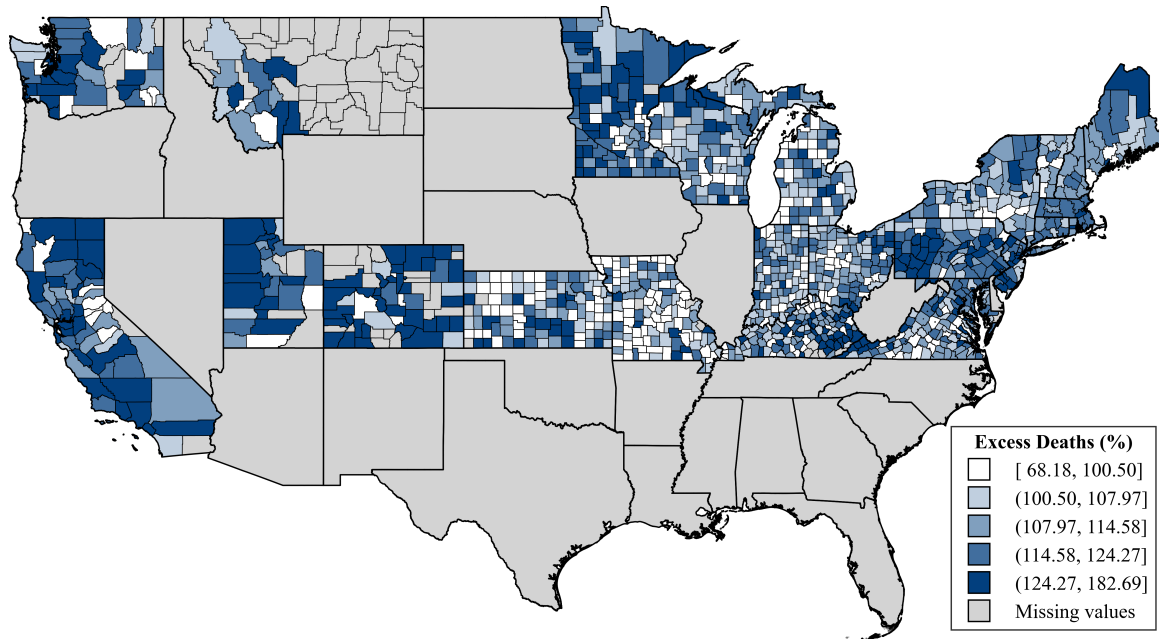
Figures

Figure 1: Name-Based Religiosity Scores



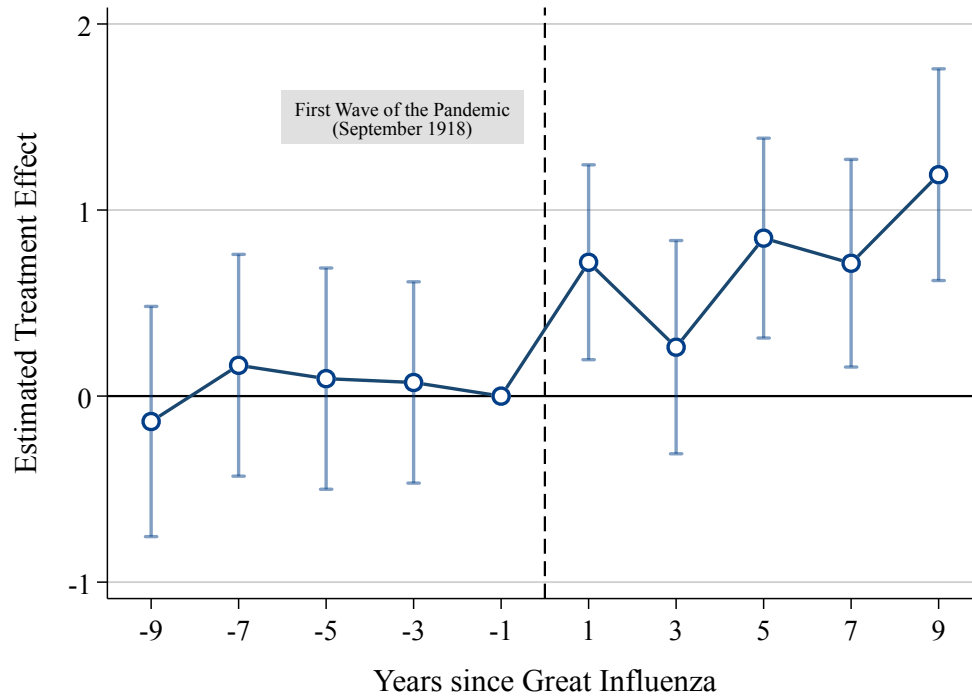
Notes: This figure displays the religiosity scores estimated using equation (1). The sample includes data from the 1906–1916 Censuses of Religious Bodies and individuals born in the United States between 1896 and 1916 from the U.S. Census. We estimate religiosity scores for names appearing in at least 0.3% of the sample. Coefficients are reported in increasing order.

Figure 2: Spatial Distribution of Excess Mortality During the Great Influenza Pandemic



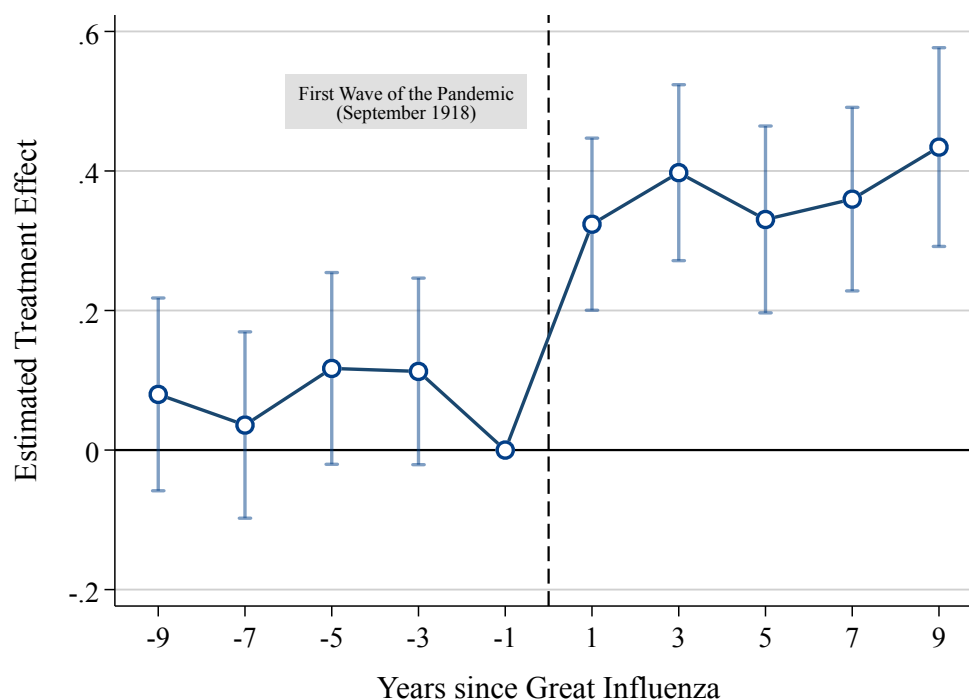
Notes: This figure displays geographic variation in excess deaths, defined in equation (3). Excess mortality is the ratio between the average yearly number of deaths during the pandemic years (1918–1919) and the average yearly number of deaths in the three years before the pandemic (1915–1917). Mortality statistics before 1915 are not available. Excess mortality is shown in percentage terms. Lighter to darker blue indicates increasing excess deaths. Counties are displayed at their 1920 borders. Mortality data are not available for states and counties displayed in gray, which are therefore excluded from the analysis.

Figure 3: Impact of the Pandemic on Religiosity



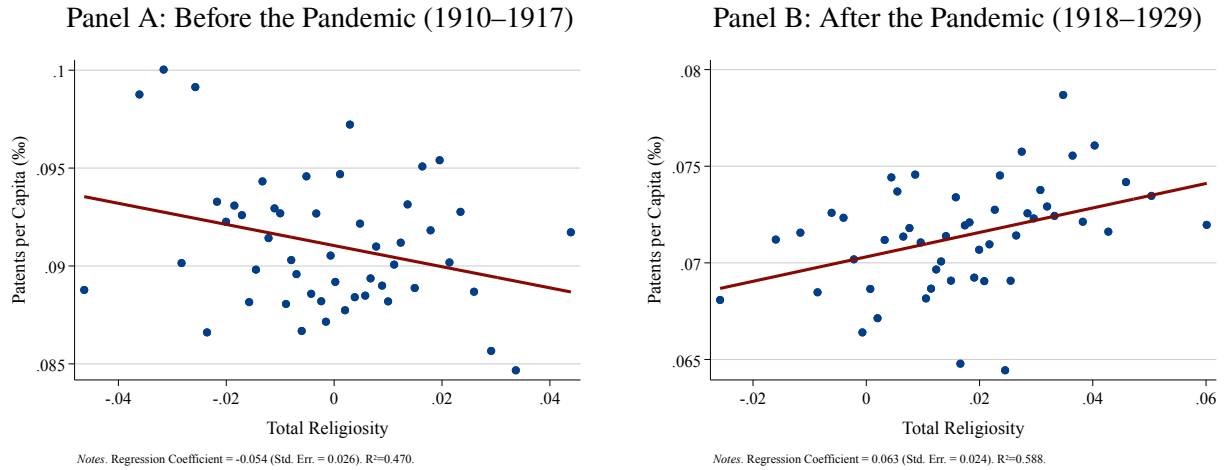
Notes: This figure displays the dynamic treatment effects of the pandemic on overall religiosity. The unit of observation is a county observed at a biennial frequency. Each dot reports the coefficient of an interaction between the baseline measure of excess deaths, defined in equation (3), and a biennial time dummy. The coefficient for 1917–1918 serves as the baseline category. The specification includes county and state-by-year-fixed effects and controls for an interaction term between the population in 1910 and a post-treatment indicator. Bands report 90% confidence intervals. Standard errors are clustered at the county level. The dashed vertical line indicates the timing of the first wave of the pandemic (September 1918).

Figure 4: Impact of the Pandemic on Innovation



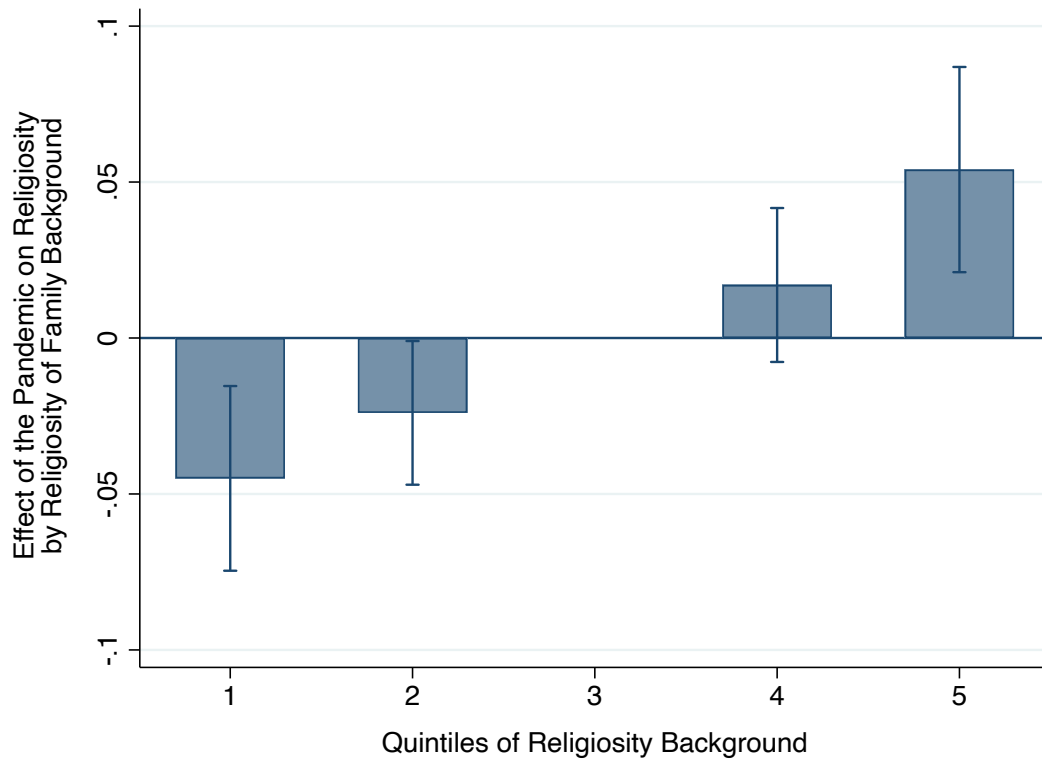
Notes: This figure displays the dynamic treatment effects of the pandemic on innovation. The dependent variable is the $\log(1 + \text{total number of patents})$ filed in a given year. The unit of observation is a county observed at a biennial frequency. Each dot reports the coefficient of an interaction between the baseline measure of excess deaths and a biennial time dummy. The coefficient for the biennial 1917–1918 serves as the baseline category. The model includes county and state-by-year-fixed effects and controls for an interaction term between the population in 1910 and a post-treatment indicator. Bands report 90% confidence intervals. Standard errors are clustered at the county level. The dashed vertical line indicates the timing of the first wave of the pandemic (September 1918).

Figure 5: Correlation Between Religiosity and Science



Notes: These figures display county-level binned scatter plots reporting the correlation between science (measured as patenting activity divided by population) and religiosity. The patents per capita variable is winsorized at the top and bottom 1%. The unit of observation is a county observed at yearly frequency. Religiosity is defined as described in section 3.1. The estimates in both plots are residualized and shown after accounting for county and year fixed effects. We report regression coefficients with standard errors and associated R^2 separately for each plot.

Figure 6: Impact of the Pandemic on the Polarization of Religious Beliefs



Notes: This figure displays the estimated impact of the pandemic on the polarization of religious beliefs by quintiles of background religiosity. Each bar reports the coefficient of an interaction between the baseline measure of excess deaths, a post-treatment indicator, and an indicator for the quintile of background religiosity. The unit of observation is a child born between 1900 and 1929. Treated children are those born after the influenza, i.e., after 1918. The dependent variable is the religiosity score associated with the child's name. Background religiosity is measured as the religiosity score of the child's father. The third quintile of background religiosity serves as the baseline category. The specification includes individual fixed effects at the household-head level: race, urban status, and place of birth. It also includes county-by-cohort, county-by-quintile of religious background, and cohort-by-quintile of religious background fixed effects. Standard errors are clustered at the county level, and the bands report the 90% confidence interval for each coefficient.

Tables

Table 1: Impact of the Pandemic on Religiosity

	Share of Affiliated			Name-Based Religiosity		
	(1)	(2)	(3)	(4)	(5)	(6)
	All	Catholics	Protestants	All	Catholics	Protestants
Excess Deaths $\times$ Post	0.220*** (0.025)	0.078*** (0.013)	0.094*** (0.017)	0.863*** (0.199)	0.269** (0.111)	0.688*** (0.171)
County FE	Yes	Yes	Yes	Yes	Yes	Yes
State-Decade FE	Yes	Yes	Yes	–	–	–
State-Year FE	–	–	–	Yes	Yes	Yes
Number of Counties	1,275	1,275	1,275	1,274	1,274	1,274
Observations	3,825	3,825	3,825	38,220	38,220	38,220
R ²	0.851	0.904	0.929	0.648	0.522	0.678
Std. Beta Coef.	0.678	0.327	0.335	0.157	0.112	0.159

Notes: This table displays the impact of exposure to the Great Influenza Pandemic on religiosity. The unit of observation is a county observed at decade frequency between 1906 and 1926 (in columns 1–3) and yearly frequency between 1900 and 1929 (in columns 4–6). “Post” is a categorical variable equal to one during and after the pandemic—i.e., in or after 1918—and zero otherwise. The baseline treatment “Excess Deaths” is defined in equation (3). In columns (1–3), the dependent variable is the number of individuals affiliated with religious denominations enumerated in the Census of Religious Bodies, divided by county population in 1900; in columns (4–6), the dependent variable is the name-based religiosity measure described in the main text. Columns (1) and (4) report the effect of the pandemic on overall religiosity. Columns (2) and (5) report the effect of the pandemic on Catholicism. Columns (3) and (6) report the effect of the pandemic on Protestantism. All columns include county fixed effects and control for an interaction term between population in 1910 and a post-treatment indicator. Columns (1) to (3) include state-by-decade fixed effects, while columns (4) to (6) include state-by-year fixed effects. Standard errors, clustered at the county level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 2: Impact of the Pandemic on Employment in Scientific Occupations

	Whole Population			Skilled Population		
	(1)	(2)	(3)	(4)	(5)	(6)
	Share STEM	Share Medical	Share Non-Medical	Share STEM	Share Medical	Share Non-Medical
Excess Deaths $\times$ Post	0.772*** (0.101)	0.217*** (0.030)	0.555*** (0.087)	12.627*** (2.416)	3.669*** (0.840)	8.957*** (1.821)
County FE	Yes	Yes	Yes	Yes	Yes	Yes
State-Decade FE	Yes	Yes	Yes	Yes	Yes	Yes
Number of Counties	1,274	1,274	1,274	1,274	1,274	1,274
Observations	3,822	3,822	3,822	3,822	3,822	3,822
R ²	0.763	0.699	0.771	0.539	0.552	0.580
Std. Beta Coef.	1.616	1.279	1.802	1.558	1.179	1.795

Notes: This table displays the impact of exposure to the Great Influenza Pandemic on employment in scientific occupations. The unit of observation is a county, observed at decade frequency between 1900 and 1930. “Post” is a categorical variable equal to one during and after the pandemic—i.e., in or after 1918—and zero otherwise. The baseline treatment “Excess Deaths” is defined in equation (3). In columns (1) and (4), the dependent variable is the share of individuals employed in STEM occupations. In columns (2) and (5), the dependent variable is the share of individuals employed in medical occupations. In columns (3) and (6), the dependent variable is the share of individuals employed in a STEM non-medical occupation. In columns (1) to (3), the share is computed with respect to the full population. In columns (4) to (6), we restrict the denominator to include only individuals employed in skilled occupations. All specifications include county fixed effects, state-by-year-fixed effects, and control for an interaction term between population in 1910 and a post-treatment indicator. Standard errors, clustered at the county level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 3: Impact of the Pandemic on Innovation

	Extensive Margin			Intensive Margin		
	(1)	(2)	(3)	(4)	(5)	(6)
	All Patents	Pharmaceutical	Non-Pharmaceutical	All Patents	Pharmaceutical	Non-Pharmaceutical
Excess Deaths $\times$ Post	0.370*** (0.056)	0.236*** (0.041)	0.319*** (0.051)	0.650*** (0.210)	0.711*** (0.224)	0.593*** (0.208)
County FE	Yes	Yes	Yes	Yes	Yes	Yes
State-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Number of Counties	1,275	1,275	1,275	265	265	265
Observations	38,250	38,250	38,250	7,950	7,950	7,950
R ²	0.858	0.791	0.852	0.898	0.801	0.888
Std. Beta Coef.	0.347	0.612	0.329	0.228	0.528	0.222

Notes: This table displays the impact of exposure to the Great Influenza Pandemic on innovation. The unit of observation is a county observed at yearly frequency between 1900 and 1929. “Post” is a categorical variable equal to one during and after the pandemic—i.e., in or after 1918—and zero otherwise. The baseline treatment “Excess Deaths” is defined in equation (3). In columns (1) and (4), the dependent variable is the log(1+ number of granted patents). In columns (2) and (5), the dependent variable is the log(1+ number of granted pharmaceutical patents). In columns (3) and (6), the dependent variable is the log(1+ number of granted non-pharmaceutical patents). In columns (4) to (6), we restrict the sample to counties with at least one granted patent every year. All regressions include county fixed effects, state-by-year-fixed effects, and control for an interaction term between population in 1910 and a post-treatment indicator. Standard errors, clustered at the county level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 4: Impact of the Pandemic on Individual Religiosity: STEM and Non-STEM

	Whole Population	Skilled Population
	(1)	(2)
	Name-Based Religiosity	Name-Based Religiosity
Excess Deaths $\times$ Post $\times$ Employed in STEM	-0.061*** (0.023)	-0.061*** (0.023)
Individual Controls	Yes	Yes
County–Year FE	Yes	Yes
County–Employed in STEM FE	Yes	Yes
Year–Employed in STEM FE	Yes	Yes
Number of Counties	1,274	1,274
Observations	10,526,699	3,678,297
$R^2 - \text{squared}$	0.008	0.014

Notes: This table displays the impact of exposure to the pandemic on the religiosity of individuals employed in STEM occupations or in other occupations. The unit of observation is a child born between 1900 and 1929 in a household where the household head is employed in a skilled occupation. Religiosity is defined as the religiosity score associated with the child’s name. “Post” is a categorical variable equal to zero for children born before the pandemic—i.e., before 1918—and one for those born after the pandemic—i.e., in or after 1918. The baseline treatment “Excess Deaths” is defined in equation (3). “Employed in STEM” is an indicator variable equal to one if the household head is employed in a STEM occupation or zero otherwise. In all specifications, the observations are weighted by the inverse of the number of children in the household. All specifications include a vector of individual controls for the household-head level: race, urban status, and place of birth as well as STEM-by-county, STEM-by-cohort, and county-by-cohort fixed effects. The sample includes the entire population in column (1) and only individuals employed in skilled occupations—as defined in Table B.2—in column (2). Standard errors are clustered at the county level and are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 5: Impact of the Pandemic by Religious Background

	Whole Population		Skilled Population	
	(1)	(2)	(3)	(4)
	Employed in STEM	Name-Based Religiosity	Employed in STEM	Name-Based Religiosity
Excess Deaths $\times$ Post $\times$ High Religious Background	-0.701** (0.337)	0.058*** (0.012)	-1.558* (0.849)	0.045** (0.020)
Individual Controls	Yes	Yes	Yes	Yes
County–Year FE	Yes	Yes	Yes	Yes
High Religious Background–Year FE	Yes	Yes	Yes	Yes
High Religious Background–County FE	Yes	Yes	Yes	Yes
Number of Counties	1,275	1,275	1,275	1,275
Observations	5,347,751	4,246,353	2,229,754	1,566,801
R ²	0.016	0.058	0.030	0.080

Notes: This table displays the impact of exposure to the pandemic on occupational choice—columns (1) and (3)—and on religiosity—columns (2) and (4)—by individual-level background religiosity. The unit of observation in columns (1) and (3) is an adult. The dependent variable “Employed in STEM” is an indicator equal to one if an individual is employed in a STEM occupation—as defined in Table B.2—or zero otherwise. “Post” is an indicator equal to one if an individual is aged between 18 and 25 in 1918 or zero otherwise. In columns (2) and (4), the unit of observation is a head of household observed once for each child born between 1900 and 1929. “Name-Based Religiosity” is defined as the religiosity score associated with the child’s name. The observations are weighted by the inverse of the number of children in the household. “Post” is an indicator equal to zero for children born during and after the pandemic—i.e., over 1918–1929—or zero for those born before the pandemic—i.e., before 1918. The baseline treatment “Excess Deaths” is defined in equation (3). “High Background Religiosity” is an indicator variable taking value one if the religiosity score of the name of the head of the household is equal to or above the median, or zero if it is below the median. The table displays the coefficient of the interaction between these terms. All specifications include a vector of individual controls for the household head: race, urban status, and place of birth as well as county-by-year, high religious background-by-year, and high religious background-by-county fixed effects. The sample includes the entire population in columns (1)–(2) and only individuals employed in skilled occupations—as defined in Table B.2—in columns (3)–(4). Standard errors are clustered at the county level and are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Online Appendix

Dealing with Adversity: Religiosity or Science? Evidence from the Great Influenza Pandemic

Enrico Berkes, Davide M. Coluccia, Gaia Dossi, and Mara P. Squicciarini

July, 2025

A Data Appendix

This section lists the data sources and describes how we construct the variables used in the analysis.

A.1 Names

Data on names are from the individual full-count US 1930 population census (Ruggles et al., 2021). Since first name records in the census conflate first and middle names, we consider only the first word that appears in each string. The baseline analysis includes all names that appear in at least 0.3% of the population of US-born children between 1900 and 1930. We perform a robustness exercise and show that the baseline results are not sensitive to alternative frequency thresholds. We exclude first-generation immigrants because their names would not reflect exposure to the pandemic as they were not born in the United States.

A.2 Religious Affiliations

Data on religious affiliations are made available by the National Historical Geographic Information System (NHGIS) and are originally from the Census of Religious Bodies, which was administered at decade frequency between 1906 and 1936. We discard the 1936 census because previous research shows that the uptake was low and unequal across counties (Stark, 1992). Census enumerators asked churches, congregations, and other local organizations to report the number of affiliates. The data was then aggregated at the county level. In our analysis, “Total Religiosity” is computed as the simple sum of religious members across all possible denominations; “Catholics” are enumerated as such. We collectively refer to “Protestants” as a set of denominations that we manually mapped to some branch of Protestantism (including, e.g., the Methodist, Evangelical, and—various—Baptist churches).

We use data on the names of saints and biblical characters from Abramitzky et al. (2016) to develop an additional indicator of county-level religiosity. To construct this additional measure, we use the baseline equation (2) and substitute the estimated $\hat{\gamma}^k$ coefficients with three indicator variables equal to one if the given name is that of a saint, a biblical character, or either.

A.3 Patents

A.3.1 Patent Data

Patent data are from Berkes (2018), who performed optical character recognition (OCR) on original patent documents issued by the United States Patents and Trademark Office between 1836 and 2010. Information includes the filing and issue year, author name, latitude and longitude of the inventor(s), and inferred USPC technology class. The data contain a set of additional variables, including the complete text of the patent document and the issue year of the patent, not used in our analysis. We geo-code each patent to its 1920 county using boundary shapefiles from NHGIS. When we collapse

by county year, we weigh each patent by the inverse of the number of technology classes and by the inverse of the number of authors. Hence, a patent with two authors and two technological classes appears four times in the original patent-level dataset, and each instance is assigned a .25 weight when aggregating at the county level. We code USPC classes to the NBER classification (Hall et al., 2001). We modify the canonical NBER classification and conflate the “Chemical” and “Drugs” categories into a single “Pharmaceuticals” class. Since multiple USPC codes are typically assigned to a single patent, most patents that fall under “Drugs” would also appear as “Chemical.” To avoid this, we recast them into one category. It is worth noting that all the results we present regarding pharmaceutical patents also hold if we keep the “Chemical” and “Drugs” classes separate.

A.3.2 Patent Importance

We measure patent ‘importance’ using the measure developed by Kelly et al. (2021). From their data, we derive two metrics. One is the number of “Breakthrough” innovations, which are defined as any patent whose importance is in the top 20% of the overall importance distribution. The second variable is the share of breakthrough patents relative to the overall number of issued patents. Both measures are net of grant-year fixed effects. We use forward and backward similarity defined within a 5-year window around the issue year of the patent.

A.4 Occupational Structure

Individual-level data on occupations is extracted from the 1930 individual-level population census. More precisely, we use the 1950 harmonized occupation classification. We then manually map occupational codes to STEM and medical occupations as described in Table B.2.

A.5 Controls & Mortality Statistics

We extract individual-level information on race and urban-rural status from the IPUMS full-count data.

County-level covariates are from NHGIS. This aggregates individual-level data from population censuses and reports data from manufacturing and agricultural censuses. All data come at historical county borders.

Mortality statistics are likewise provided by NHGIS. For the period we are interested in, 1915-1919, they were collected for about 1,200 counties, covering approximately 60% of the US population. We measure Influenza-related mortality as the ratio between average yearly deaths in 1918 and 1919, and average yearly deaths in 1915 to 1917.⁴¹

⁴¹The original documents report, for major cities, deaths broken down by (alleged) cause. We do not use this data for two main reasons. First, they are incomplete and are only available for cities. Second, Beach et al. (2022) criticizes the methodology adopted to impute the cause of deaths.

A.6 Other Data

In several robustness regressions, we control for WW1 mortality. The underlying data are from Ferrara and Fishback (2024) and were kindly shared by the authors.

A.7 Boundary Harmonization

County-level data from NHGIS and other sources are typically provided at historical borders. To ensure comparability and consistency, we adopt the methodology developed by Eckert, Gvirtz, Liang, and Peters (2018) to compute geographical crosswalks between US counties over time. In a nutshell, their methodology is as follows. Suppose we know the distribution of a given variable y across counties at decade frequency between 1900 and 1930. To harmonize borders to one year, Eckert et al. (2018) overlay the shapefile of counties in a given year, say, 1900, to that in the reference year, say, 1920. They then compute the percentage of land that a given county shares with itself between the two years and that assigned to other counties. To construct the harmonized variable, one multiplies these overlapping area weights by the variable recorded in 1900 and aggregates up by 1920 counties. The underlying assumption is that y is evenly distributed over the county territory. While this assumption may seem too demanding in some cases, departures from it are plausibly innocuous in our setting. First, county borders had undergone major consolidations before 1900 and remained stable thereafter. Moreover, mortality data are mostly available for the Northwest and Midwest areas, where boundary changes were rare and minor after the 1890s. In our analysis, we map all county-level variables to 1920-borders.

A.8 Details on Sample Construction

In this paragraph, we provide additional technical details on how we construct the estimation samples. The main sample restriction comes from the fact that we observe mortality for 1,302 out of 3,100 counties. We then discard 27 counties with values of excess mortality below or above, respectively, the bottom 1% (85% of the pre-pandemic mortality) and top 1% (180% of the pre-pandemic level) of the excess-mortality distribution. Because such figures are due to scarcely-inhabited areas, these 27 counties account for less than 0.1% of the population in the 1,302 counties sample. We are left with 1,275 counties. In the rest of the paragraph, we explain why we may not always be able to use all 1,275 for the estimation.

A.8.1 County-Level Religiosity

The yearly county religiosity estimation sample is a balanced panel where each county is observed at a yearly frequency between 1900 and 1929. This implies that the number of counties in this balanced panel may not be 1,275 if at least one county is not observed at least once between 1900 and 1929. This happens because, especially in scarcely-inhabited areas, our name-frequency threshold may imply that we cannot match any newborn in a given cohort. If that is the case, the county will subsequently be

dropped from the estimation sample. This happens for one county, so the estimation sample for the county-level religiosity analysis consists of 1,274 counties accounting for 99.5% of the population in the 1,302-county sample.

In the decade-level county panel, counties are observed at decade frequency instead. In this case, the sample is constructed from adults observed once per census decade between 1900 and 1930, and the post-treatment indicator returns a value of one for decades 1920 and 1930 and zero otherwise. Religiosity is constructed from the Census of Religious Bodies, as the number of people affiliated to a denomination, normalized by population in 1910, i.e., the last year before the Influenza.

A.8.2 County-Level Innovation

The yearly county-level innovation sample is a balanced panel where each county is observed at a yearly frequency between 1900 and 1929. Thus, an observation in the dataset can either be a number above zero (if one or more patents are observed in that county year) or zero (if no patents are observed). The estimation sample in this case includes all 1,275 counties for which we observe mortality. In columns (2) and (7) of Table B.13 we don't include county-year when no patents are observed. This results in an unbalanced panel dataset where a county may not be observed every year over the estimation period.

The decade county-level innovation sample is a balanced panel where each unit is observed at a decade frequency. In this case, the sample is constructed from adults observed once per census decade between 1900 and 1930. The post-treatment variable is an indicator equal to one for decades 1920 and 1930 and zero otherwise. The dependent variable is the share of individuals employed in a STEM occupation (listed in Table B.2), normalized by population in 1910, i.e., the last census before the Influenza.

A.8.3 Individual-Level

We construct two individual-level datasets. The first sample ("adult sample") comprises all those born before 1900. Each individual is observed once in the 1930 census. Because the adult sample is used to study the evolution of the occupational structure, we discard (i) individuals with missing occupational response, (ii) farmers, who have been shown to display disproportionately high occupational persistence (Long and Ferrie, 2013), and (iii) women, since female labor force participation was extremely low. The second sample ("children sample") comprises all those born between 1900 and 1930. Each household is observed once per child. In the children sample, we excluded households when (i) the father's occupation was missing and (ii) there was no father since here, too, and the labor force participation of women was very low. We identify a household as "STEM" if at least one of its members is employed in a STEM occupation. In addition, we excluded a small fraction of households ($\approx 0.7\%$) where we cannot unambiguously identify the father. Consistently with the county-level analysis, we

do not assign a religiosity score to first-generation immigrants.

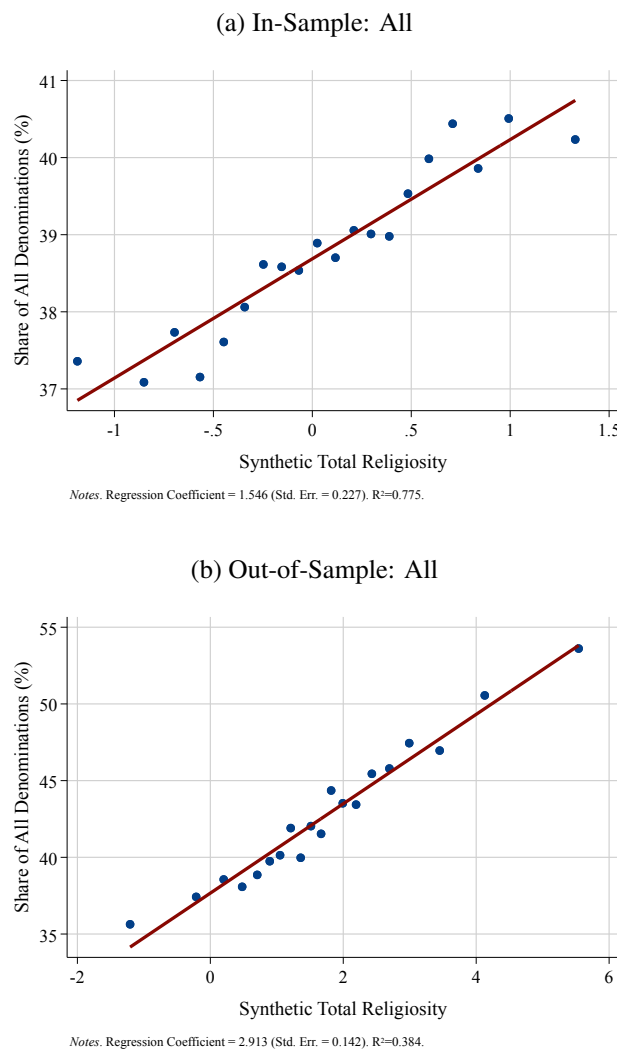
A.8.4 City-Level

To build the city-level sample, we construct the baseline excess deaths treatment variable from data by Clay et al. (2019). The dataset contains information for 982 cities. We observe mortality continuously between 1915 and 1919—the period over which we compute excess mortality in the county-level sample—for a subset of 493 cities. Moreover, we do not observe population data in 1900 for four additional cities, and, in 44 cases, we do not observe newborn children in each year of the sample period. The final sample of 489 cities is thus reduced to 445 for religiosity regressions. As in the county-level sample, we exclude cities in the top and bottom 1% of the excess mortality distribution. In Figure B.5, we report the location of each of the 489 cities and the number of cities included in the sample by state. We geo-code patents to historical city borders and construct the name-based religiosity measure from the individuals recorded living in each city in the 1930 census. The city-level sample is used in the regressions shown in Table B.9.

B Additional Figures and Tables

B.1 Appendix Figures

Figure B.1: In-Sample and Out-of-Sample Fit of Name-Based Religiosity Measure



Notes: These figures display county-level binned scatter plots reporting the correlation between our religiosity measure and the number of affiliated members to all denominations (B.1a-B.1b) divided by population in 1900. In-sample figures report data for the 1906 and 1916 censuses of religious affiliations. Out-of-sample figures instead report data for 1926. In-sample regressions include county-fixed effects; out-of-sample regressions include state-fixed effects. In the note, we report the regression coefficients and the associated R^2 .

Figure B.2: Example of Pharmaceutical Patent

(a) Text

Patented Mar. 1, 1927.

1,619,005

UNITED STATES PATENT OFFICE.

SAMUEL M. STRONG, OF GARDEN CITY, NEW YORK.

RESPIRATION AND PULSE RECORDER.

Application filed January 11, 1922. Serial No. 528,485.

This invention relates to a device or instrument for recording the character of the actions of the heart and respiratory organs of a person.

The primary object of the invention is to provide an instrument which will produce an accurate graphic representation of the rate, rhythm, and force of respiration and pulse of a human being over a short or a long period of time.

ing plate 13 and a vertical portion 19 adapted to be placed against a side plate of the casing 10. The main bearing plate fits snugly within the casing and one end of the horizontal portion 18 abuts against the cover 11 when the latter is in position. A side bearing plate 20 is located immediately adjacent to the detachable side plate 11 and an intermediate bearing plate 21 is interposed between the bearing plate 20 and the

(b) Figures

March 1, 1927.

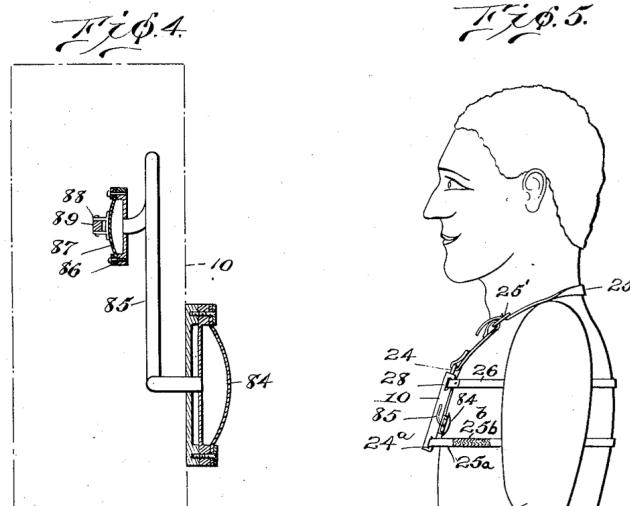
S. M. STRONG

1,619,005

RESPIRATION AND PULSE RECORDER

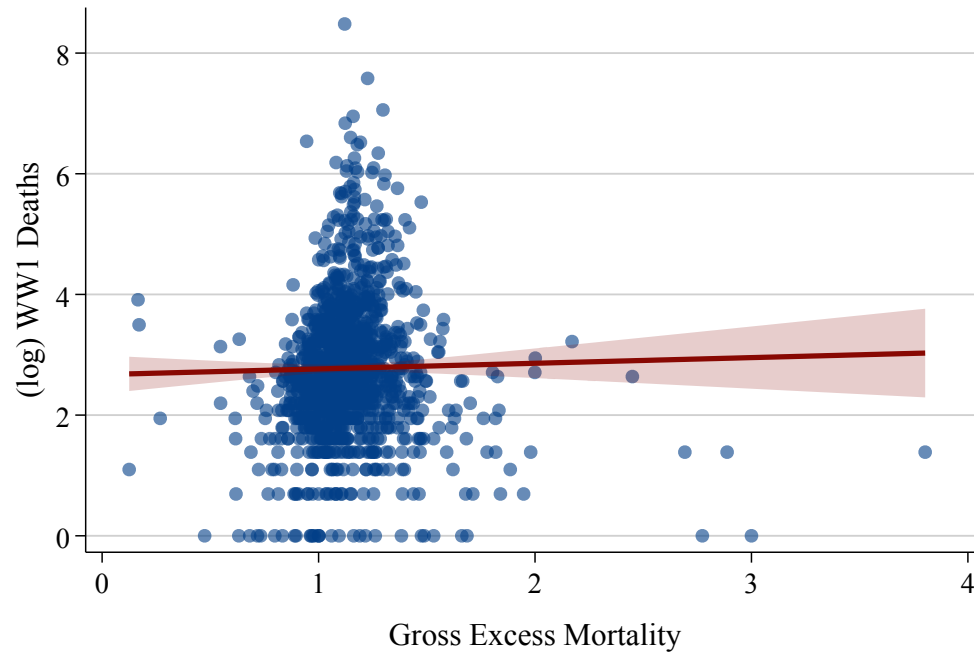
Filed Jan. 11, 1922

2 Sheets-Sheet 2



Notes: This figure displays the text and figures of a patent classified as pharmaceutical.

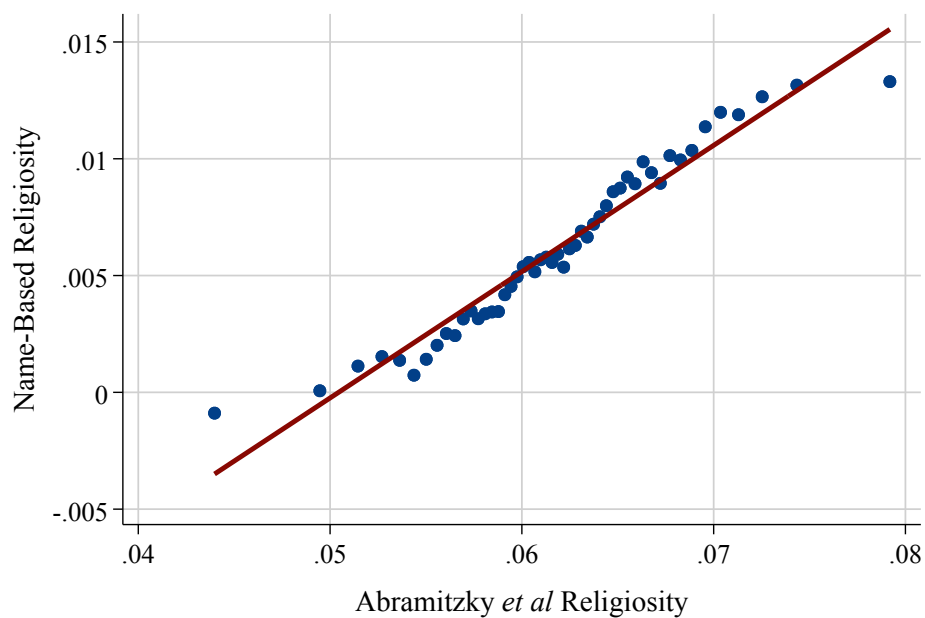
Figure B.3: Correlation Between WW1 Deaths and Excess Deaths



Notes. Regression Coefficient = 0.094 (Std. Err. = 0.140). $R^2=0.000$.

Notes: This figure displays the correlation between WW1 deaths (in log) and excess mortality due to the Influenza pandemic. In the notes, we report the regression coefficient and the R^2 . Data on WW1 deaths are from Ferrara and Fishback (2024).

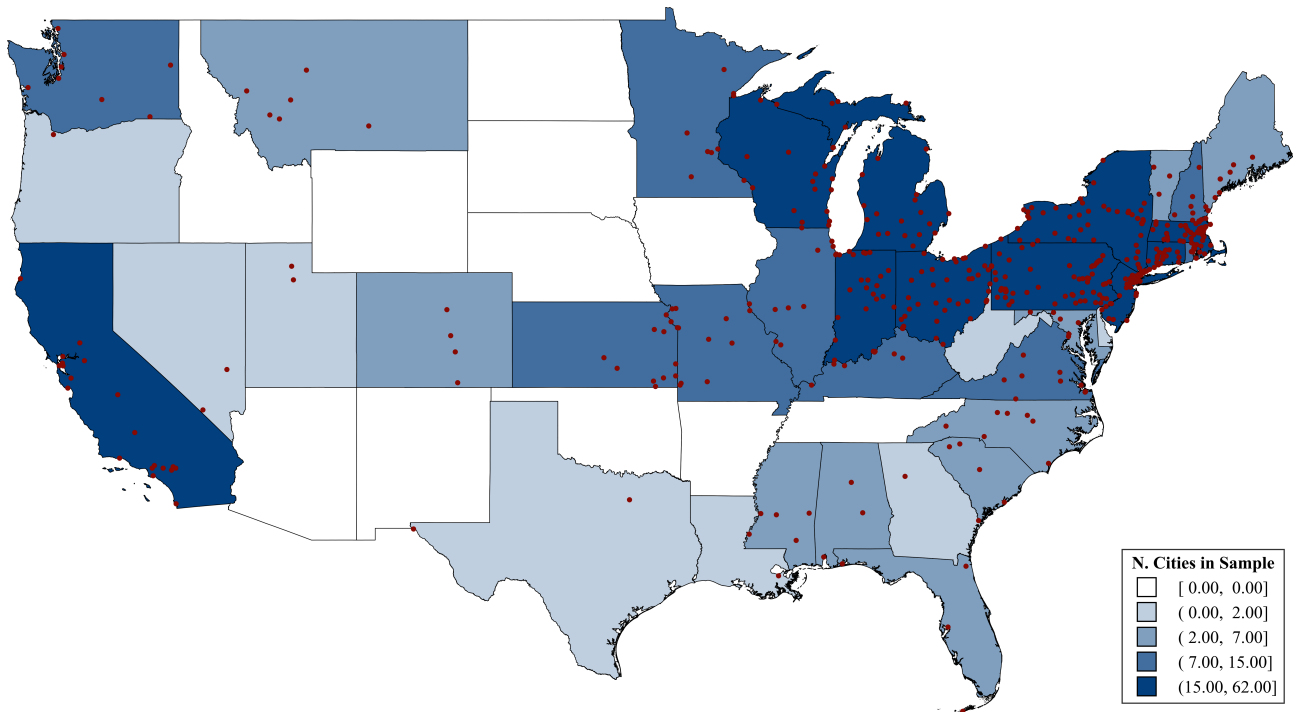
Figure B.4: Correlation Between Abramitzky et al. (2016) Religiosity and Baseline Religiosity



Notes. Regression Coefficient = 0.541 (Std. Err. = 0.018). $R^2=0.584$.

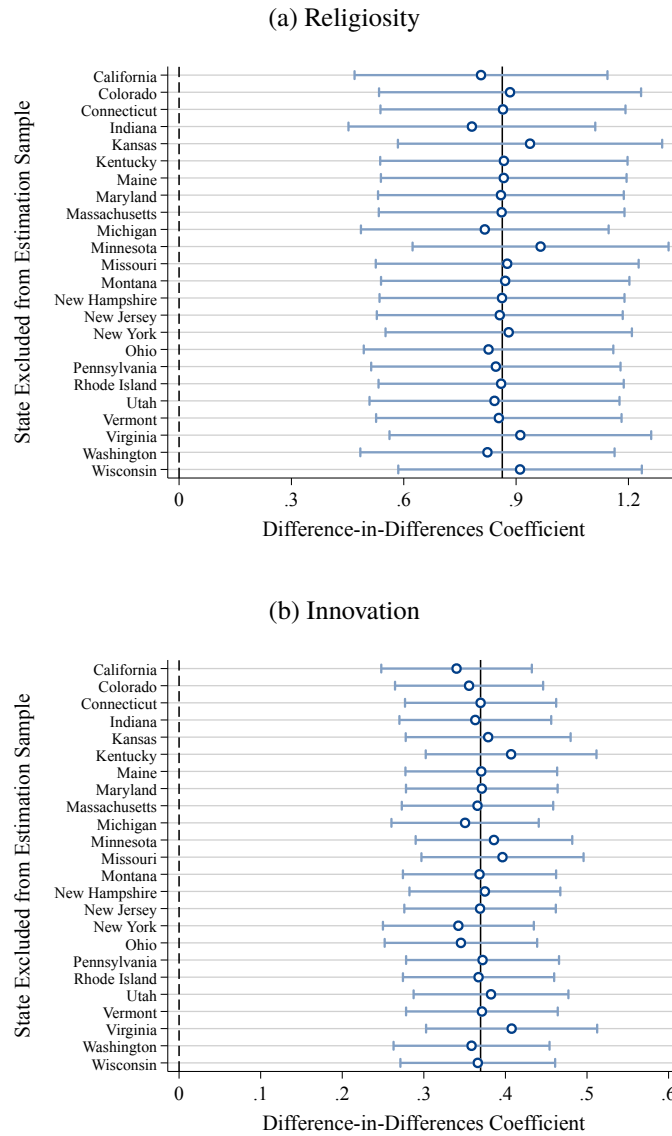
Notes: This figure displays the correlation between our baseline religiosity measure (multiplied by 100) and the share of biblical or saint names, as defined in Abramitzky et al. (2016). The unit of observation is a county observed at yearly frequency between 1900 and 1930. The graph partials out county and year fixed effects. The notes report the regression coefficient and the associated standard error, clustered at the county level, and R^2 .

Figure B.5: Distribution of Cities in the City-level Sample



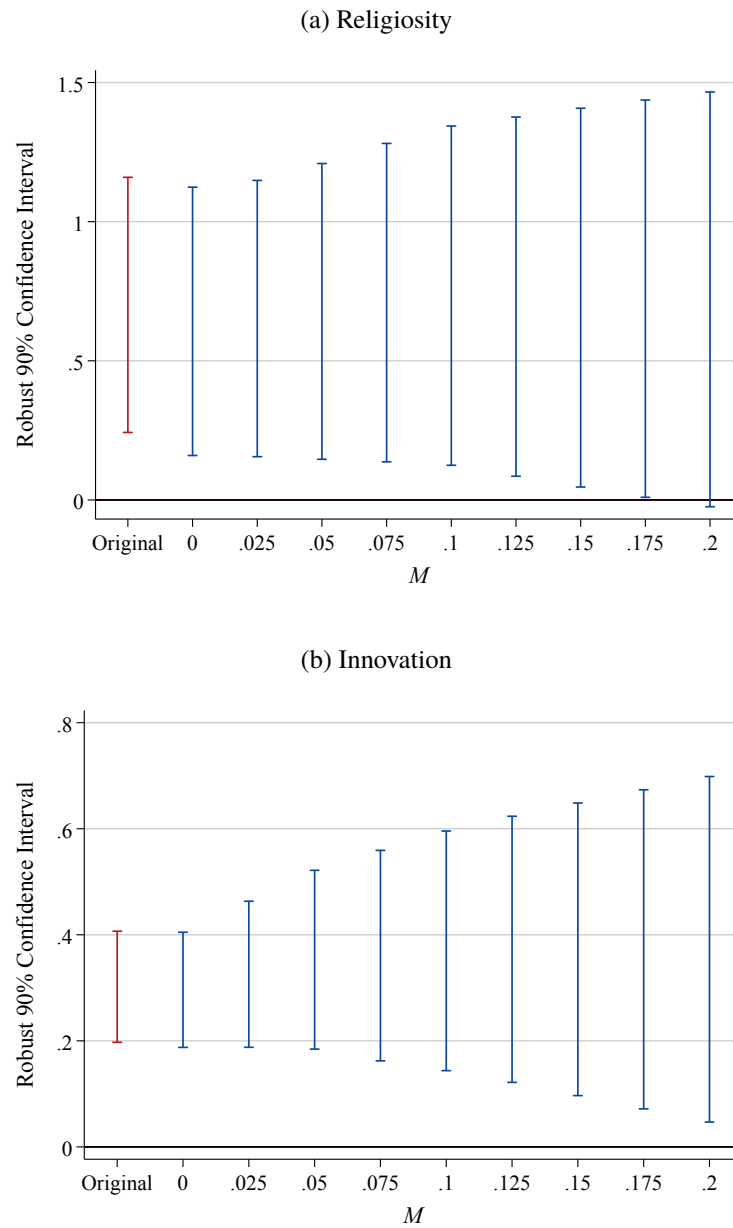
Notes: This figure displays the spatial distribution of the cities in the city-level sample used in Table B.9. We use data from Clay et al. (2019). The red dots report the coordinates of the 489 cities for which we can construct the excess mortality treatment measure. Lighter to darker shades of blue indicate the state-level number of cities included in the final sample.

Figure B.6: County-Level Estimates Excluding One State at a Time



Notes: This table displays the impact of exposure to the Great Influenza Pandemic on religiosity and innovation. The unit of observation is a county observed at yearly frequency between 1900 and 1929. Each dot reports the coefficient of an interaction between a “Post” categorical variable equal to one during and after the pandemic—i.e., in or after 1918—or zero otherwise and the baseline treatment “Excess Deaths” defined in equation (3), dropping one state at a time from the estimation sample. In Panel B.6a, the dependent variable is the name-based religiosity measure; in Panel B.6b, the dependent variable is the log(1+ number of granted patents). All regressions include county and state-by-year fixed effects and control for an interaction term between population in 1910 and a post-treatment indicator. Standard errors are clustered at the county level, and the bands report 90% confidence intervals. The solid black line reports the treatment effect on the full sample.

Figure B.7: Sensitivity Analysis of Confidence Intervals for Non-Linear Pre-Trends



Notes: This figure displays sensitivity analyses of the estimated effect of the pandemic on religiosity (Panel B.7a) and innovation (Panel B.7a) to potential violations of parallel trends assumption, following Rambachan and Roth (2023). The red bar represents the 90% confidence interval of the first post-pandemic event study estimate. The blue bars report corresponding 90% robust confidence intervals that allow for per-period violations of parallel trends up to M . Setting M equal to 0 corresponds to a linear differential trend.

B.2 Appendix Tables

Table B.1: Summary Statistics

	(1)	(2)	(3)	(4)	(5)
	Mean	Std. Dev	Min	Max	Counties
Panel A. Mortality					
Flu Excess Deaths (%)	1.127	0.163	0.682	1.827	1275
WW1 Deaths	39.834	167.083	0.000	4828.000	1275
Panel B. Religion					
All Denominations - Census Based	40.525	14.099	0.000	94.264	1275
Catholics - Census Based	11.024	12.007	0.000	65.853	1275
Protestants - Census Based	27.448	13.807	0.000	94.264	1275
All Denominations - Name Based	0.016	0.021	-0.039	0.103	1274
Catholics - Name Based	0.001	0.008	-0.030	0.045	1274
Protestants - Name Based	0.014	0.017	-0.027	0.084	1274
Panel C. Scientific Progress					
STEM Employment Share	0.480	0.313	0.054	2.334	1275
Medical Employment Share	0.170	0.093	0.022	1.388	1275
STEM Non-Medical Employment Share	0.310	0.262	0.000	1.912	1275
Total	278.976	1095.265	0.000	15000.000	1275
Pharmaceutical	49.756	200.266	0.000	2708.315	1275
Communication	11.166	54.019	0.000	1110.340	1275
Electrical	40.552	196.034	0.000	4005.376	1275
Mechanical	136.779	533.283	0.000	7834.571	1275
Other	149.124	568.327	0.000	7290.446	1275
Panel D. Income and Demographics					
Population	130.432	329.177	1.061	4819.392	1275
Area	216.975	280.512	0.277	5205.795	1275
Occupational Score per Capita	835.998	74.425	311.939	2302.648	1275
Share of Men	52.305	4.856	19.177	131.582	1275
Share of Illiterates	72.620	9.326	24.633	196.336	1275
Share of Young	33.241	5.981	12.927	86.595	1275
Share of Whites	94.634	11.952	26.163	100.000	1275
Share of African Americans	5.366	11.952	0.000	73.837	1275
Share of Foreign Born	10.680	9.928	0.008	44.052	1275

Notes: This table displays mean, standard deviation, minimum, maximum, and total number of counties for the variables used in the main analysis. Observations are at the county level. Influenza mortality in Panel A is constructed from the mortality statistics; WW1 deaths are from Ferrara and Fishback (2024). Data in Panel B are from either the Census of Religious Bodies at the decade level or are constructed from name frequencies at the year level. Patent data in Panel C are from Berkes (2018) and are aggregated at the decade level, while scientific employment is tabulated from the IPUMS full-count 1910 census (Ruggles et al., 2021). The share of STEM is computed from the census and is in 1,000 units. Panel D reports data from the 1910 census. County demographics are measured through the IPUMS full-count census (Ruggles et al., 2021). Panel B data and Panel D population are expressed in 1,000 units. All variables are crosswalked to 1920 borders.

Table B.2: Medical, STEM, and Skilled Occupations

Code (1)	Occupation Label (2)	Code (3)	Label (4)
<i>Panel A. Medical Occupations</i>			
19	Medical Sciences	59	Nurses, Student Professional
32	Dentists	75	Physicians and Surgeons
58	Nurses, Professional		
<i>Panel B. STEM Occupations (Includes Medical Occupations)</i>			
7	Chemists	48	Engineering, Mining
12	Agricultural Sciences	49	Engineering (n.e.c.)
13	Biological Sciences	58	Nurses, Professional
14	Chemistry	59	Nurses, Student Professional
16	Engineering	61	Agricultural Scientists
17	Geology and Geophysics	62	Biological Scientists
18	Mathematics	63	Geologists and Geophysicists
19	Medical Sciences	67	Mathematicians
23	Physics	68	Physicists
25	Statistics	69	Miscellaneous Natural Scientists
26	Natural Sciences (n.e.c.)	71	Osteopaths
32	Dentists	75	Physicians and Surgeons
34	Dietitians and Nutritionists	83	Statisticians and Actuaries
41	Engineering, Aeronautical	92	Surveyors
42	Engineering, Chemical	98	Veterinarians
43	Engineering, Civil	240	Officers, Pilots, Purser, and Engineers, Ships
44	Engineering, Electrical	541	Locomotive Engineers
45	Engineering, Industrial	563	Opticians and lens grinders and polishers
46	Engineering, Mechanical	583	Stationery Engineers
47	Engineering, Metallurgical, Metallurgists		
<i>Panel C. Skilled Occupations (Includes STEM and Medical Occupations)</i>			
$1 \leq \cdot \leq 99$	Professional, Technical	$200 \leq \cdot \leq 299$	Managers, Officials, and Proprietors
$500 \leq \cdot \leq 595$	Craftsmen		

Notes: Panel A lists occupations that we classify as “Medical.” Panel B lists occupations that we classify as “Science, Technology, Engineering, and Mathematics” (STEM). Panel C lists occupations that we classify as “Skilled.” Occupation codes and occupation labels are from the IPUMS harmonized 1950 occupation taxonomy (variable “OCC1950”).

Table B.3: Balance Checks: Regressions

	(1)	(2)	(3)
	Coefficient	Standard Error	95% C. I.
<i>Panel A. Religion</i>			
All Denominations (Census Based)	0.057	(0.176)	[-0.245, 0.359]
Catholics (Census Based)	0.168	(0.147)	[-0.084, 0.421]
Protestants (Census Based)	-0.094	(0.149)	[-0.349, 0.161]
All Denominations (Name Based)	0.211	(0.224)	[-0.173, 0.595]
Catholics (Name Based)	-0.079	(0.196)	[-0.415, 0.258]
Protestants (Name Based)	0.350*	(0.191)	[-0.022, 0.678]
<i>Panel B. Scientific Progress</i>			
STEM Employment Share	0.389	(0.281)	[-0.093, 0.870]
Medical Employment Share	0.248	(0.162)	[-0.030, 0.526]
Non-Medical Employment Share	0.377	(0.287)	[-0.115, 0.869]
Total Patents	0.063	(0.055)	[-0.031, 0.158]
Pharmaceutical Patents	0.063	(0.059)	[-0.038, 0.165]
Non-Pharmaceutical Patents	0.063	(0.054)	[-0.030, 0.157]
<i>Panel C. Income and Demographics</i>			
Population Density	-0.009	(0.206)	[-0.363, 0.344]
Share of Men	0.084	(0.056)	[-0.011, 0.179]
Share of Illiterates	-0.147	(0.138)	[-0.384, 0.089]
Share of Young	0.097	(0.146)	[-0.154, 0.348]
<i>Panel D. Ethnic Composition</i>			
Share of Whites	0.096	(0.092)	[-0.062, 0.253]
Share of African Americans	-0.096	(0.092)	[-0.253, 0.062]
Share of Foreign Population	0.310***	(0.110)	[0.122, 0.498]
Immigrants from:			
Italy	0.191	(0.138)	[-0.045, 0.427]
Ireland	-0.141	(0.123)	[-0.351, 0.070]
Austria	0.275**	(0.116)	[0.075, 0.474]
France	-0.044	(0.074)	[-0.171, 0.830]
Spain	-0.007	(0.055)	[-0.101, 0.087]
Portugal	-0.063	(0.157)	[-0.333, 0.207]

Notes: This table displays the correlation between excess deaths defined in equation (3) and a set of covariates in 1910, the last census year before the pandemic. Column (1) reports the standardized coefficient of a regression between the row variable and our measure of excess deaths; column (2) reports the associated standard error in round brackets; column (3) reports the confidence interval of the point estimate at 90% confidence level in square brackets. All variables are expressed as shares of total population, except for population density. Regressions control for county population and include state-fixed effects. Standard errors are clustered at the county level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.4: Impact of the Pandemic on Religiosity: Weighted Regressions

	Share of Affiliated			Name-Based Religiosity		
	(1)	(2)	(3)	(4)	(5)	(6)
	All	Catholics	Protestants	All	Catholics	Protestants
Excess Deaths $\times$ Post	0.264** (0.135)	0.119** (0.056)	0.137*** (0.029)	1.052*** (0.274)	0.298 (0.247)	1.084*** (0.229)
County FE	Yes	Yes	Yes	Yes	Yes	Yes
State-Decade FE	Yes	Yes	Yes	–	–	–
State-Year FE	–	–	–	Yes	Yes	Yes
Number of Counties	1,275	1,275	1,275	1,274	1,274	1,274
Observations	3,825	3,825	3,825	38,220	38,220	38,220
R ²	0.775	0.888	0.925	0.843	0.768	0.844
Std. Beta Coef.	0.696	0.398	0.626	0.165	0.108	0.228

Notes: This table displays the impact of exposure to the Great Influenza Pandemic on religiosity. The unit of observation is a county observed at decade frequency between 1906 and 1926 (in columns 1–3) and yearly frequency between 1900 and 1929 (in columns 4–6). “Post” is a categorical variable equal to one during and after the pandemic—i.e., in or after 1918—and zero otherwise. The baseline treatment “Excess Deaths” is defined in equation (3). In columns (1–3), the dependent variable is the number of individuals affiliated with religious denominations enumerated in the Census of Religious Bodies, divided by county population in 1900; in columns (4–6), the dependent variable is the name-based religiosity measure described in the main text. Columns (1) and (4) report the impact of the pandemic on overall religiosity. Columns (2) and (5) report the impact of the pandemic on Catholicism. Columns (3) and (6) report the impact of the pandemic on Protestantism. All columns include county fixed effects and control for an interaction term between population in 1910 and a post-treatment indicator. Columns (1) to (3) include state-by-decade fixed effects, while columns (4) to (6) include state-by-year fixed effects. Counties are weighted by population in 1910. Standard errors, clustered at the county level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.5: Impact of the Pandemic on Religiosity: Additional Checks

	(1)	(2)	(3)	(4)
	Name-Based Religiosity			
Excess Deaths Dummy $\times$ Post	0.252*** (0.060)			
Excess Deaths $\times$ Post		0.824*** (0.199)	0.876*** (0.199)	0.001 (0.002)
WWI Deaths $\times$ Post		-0.000 (0.001)		
County FE	Yes	Yes	Yes	Yes
Year–State FE	Yes	Yes	Yes	No
Decade–State FE	–	–	–	Yes
Sample	Baseline	Baseline	No Internal Migrants	Adults
Number of Counties	1,274	1,274	1,274	1,267
Observations	38,220	38,220	38,220	5,068
R ²	0.648	0.648	0.646	0.782
Std. Beta Coef.	0.033	0.150	0.161	0.028

Notes: This table displays the impact of exposure to the Influenza on overall religiosity. The unit of observation is a county observed at yearly frequency between 1900 and 1929 in columns (1)–(3) and at decade frequency in column (4). “Post” is a categorical variable equal to one during and after the pandemic—i.e., in or after 1918 in columns (1)–(3) and the decades 1920–1930 in column (4)—and zero otherwise. The baseline treatment “Excess Deaths” is defined in equation (3). The dependent variable is the name-based measure of aggregate religiosity described in the main text. Column (1) reports the results coding the treatment as a binary variable returning value one if the continuous treatment is above its median and zero otherwise. In column (2), we control for WWI-related deaths interacted with the “Post” indicator. In column (3), we compute religiosity, excluding individuals who were not born in the state where they are recorded in the 1930 census. Column (4) reports results measuring county religiosity using the names stock of adults—which serves as a placebo check. All regressions in columns (1)–(3) include county and state-by-year-fixed effects; the regression in column (4) includes county and state-by-decade-fixed effects. Additionally, each regression controls for an interaction term between the population in 1910 and a post-treatment indicator. Standard errors are clustered at the county level and are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.6: Impact of the Pandemic on Religiosity: Alternative Sample for Name-Based Religiosity

	(1)	(2)	(3)
	Name-Based Religiosity		
Excess Deaths $\times$ Post	1.182*** (0.233)	0.916*** (0.191)	0.123*** (0.026)
County FE	Yes	Yes	Yes
Year–State FE	Yes	Yes	Yes
Sample	No Firstborn	< 5 Kids	Household
Number of Counties	1,274	1,274	1,274
Observations	38,220	38,220	38,220
R ²	0.624	0.631	0.754
Std. Beta Coef.	0.224	0.174	0.162

Notes: This table displays the impact of exposure to the Influenza on overall religiosity. The unit of observation is a county, observed at yearly frequency between 1900 and 1929. “Post” is a categorical variable equal to one during and after the pandemic—i.e., in or after 1918—and zero otherwise. The baseline treatment “Excess Deaths” is defined in equation (3). The dependent variable is the name-based measure of aggregate religiosity described in the main text. Column (1) drops first-born children in every household. In column (2), we compute religiosity dropping all children beyond the fourth in each household. In column (3), we first compute within-household average religiosity and then aggregate the resulting religiosity series at the county-year level. All regressions include county and state-by-year-fixed effects. Additionally, each regression controls for an interaction term between the population in 1910 and a post-treatment indicator. Standard errors are clustered at the county level and are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.7: Impact of the Pandemic on Religiosity: Alternative Ways of Constructing Name-Based Religiosity

	Scores w/out County FE		Alternative Thresholds (Unweighted)			Alternative Thresholds (Weighted)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Unweighted	Weighted	$(\tau = 1)$	$(\tau = 2)$	$(\tau = 4)$	$(\tau = 1)$	$(\tau = 2)$	$(\tau = 4)$
Excess Deaths $\times$ Post	2.508*** (0.878)	3.572** (1.772)	0.749*** (0.244)	0.852*** (0.238)	0.397** (0.169)	0.901** (0.375)	0.924*** (0.333)	0.390* (0.208)
County FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of Counties	1,274	1,274	1,274	1,274	1,274	1,274	1,274	1,274
Observations	38,220	38,220	38,220	38,220	38,220	38,220	38,220	38,220
R ²	0.910	0.977	0.396	0.421	0.692	0.609	0.640	0.855
Std. Beta Coef.	0.068	0.058	0.136	0.166	0.077	0.152	0.170	0.066

Notes: This table displays the impact of exposure to the Influenza on religiosity. The unit of observation is a county observed at yearly frequency between 1900 and 1929. “Post” is a categorical variable equal to one during and after the pandemic—i.e., in or after 1918—and zero otherwise. The baseline treatment “Excess Deaths” is defined in equation (3). In columns (1–2), religiosity is measured using religiosity scores obtained by estimating equation (1), except that we do not include the fixed effects in the regression specification. In column (2), counties are weighted by 1900 population. In columns (3–8), religiosity is measured using religiosity scores obtained by estimating equation (1). The term τ denotes the frequency threshold a name must exceed to be included in our sample, in ‰ terms. For instance, $\tau = 2$ implies that at least 2‰ children in our sample must be called with a given name, for that name to be included in the sub-sample of names used to compute the religiosity score. We report the estimated coefficients for different frequency threshold values in the various columns. In columns (3–5), counties are not weighted; in columns (6–8), counties are weighted by population in 1900. All specifications include county and state-by-year-fixed effects and control for an interaction term between the population in 1910 and a post-treatment indicator. Standard errors are clustered at the county level and are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.8: Impact of the Pandemic on Religiosity Measured with Saint or Biblical Names

	Religiosity Based on Saints' and Biblical Names		
	(1)	(2)	(3)
	Saints/Biblical	Saints	Biblical
Excess Deaths $\times$ Post	0.316** (0.145)	0.282** (0.136)	0.081** (0.036)
County FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Number of Counties	1,274	1,274	1,274
Observations	38,220	38,220	38,220
R ²	0.991	0.991	0.985
Std. Beta Coef.	0.028	0.026	0.026

Notes: This table displays the impact of exposure to the Influenza on religiosity. The unit of observation is a county observed at yearly frequency between 1900 and 1929. “Post” is a categorical variable equal to one during and after the pandemic—i.e., in or after 1918—and zero otherwise. The baseline treatment “Excess Deaths” is defined in equation (3). In column (1), the dependent variable is the log of the number of children by cohort whose name either appears in the bible or is the name of a saint; in column (2), the dependent variable only includes biblical names; in column (3), it only includes names of saints. Biblical and saints’ names are from Abramitzky et al. (2016). All specifications include county and state-by-year-fixed effects and control for an interaction term between the population in 1910 and a post-treatment indicator. Standard errors are clustered at the county level and are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.9: Impact of the Pandemic on Religiosity and Innovation: City-Level Analysis

	(1)	(2)
	Name-Based Religiosity	All Patents
Excess Deaths $\times$ Post	0.011*** (0.004)	0.584*** (0.210)
City FE	Yes	Yes
State-Year FE	Yes	Yes
Number of Cities	430	466
Observations	12,900	13,980
R ²	0.639	0.856
Std. Beta Coef.	0.251	0.222

Notes: This table displays the city-level effect of exposure to the Influenza on religiosity (column 1) and innovation measured by patenting activity (column 2). The unit of observation is a city observed at yearly frequency between 1900 and 1929. We report the location of each city in the sample in figure B.5. The baseline sample is from Clay et al. (2019). We include only cities where we can construct the baseline excess mortality measure. In column (1), the dependent variable is the name-based religiosity measure constructed on the universe of children born between 1900 and 1929 and residing in each city in the 1930 census; in column (2), the dependent variable is $\log(1 + \text{number of patents})$. “Post” is a categorical variable equal to one during and after the pandemic—i.e., in or after 1918—and zero otherwise. The baseline treatment “Excess Deaths” is defined in equation (3). All specifications include city and state-by-year-fixed effects. Standard errors, clustered at the city level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.10: Impact of the Pandemic on Employment in Scientific Occupations: Weighted Regressions

	Whole Population			Skilled Population		
	(1)	(2)	(3)	(4)	(5)	(6)
	Share STEM	Share Medical	Share Non-Medical	Share STEM	Share Medical	Share Non-Medical
Excess Deaths $\times$ Post	0.982*** (0.290)	0.193 (0.165)	0.789*** (0.157)	12.074*** (2.914)	2.536* (1.362)	9.538*** (1.831)
County FE	Yes	Yes	Yes	Yes	Yes	Yes
State-Decade FE	Yes	Yes	Yes	Yes	Yes	Yes
Number of Counties	1,274	1,274	1,274	1,274	1,274	1,274
Observations	3,822	3,822	3,822	3,822	3,822	3,822
R ²	0.809	0.767	0.821	0.713	0.697	0.725
Std. Beta Coef.	2.056	1.136	2.563	1.490	0.815	1.911

Notes: This table displays the impact of exposure to the Great Influenza Pandemic on employment in scientific occupations. The unit of observation is a county, observed at decade frequency between 1900 and 1930. “Post” is a categorical variable equal to one during and after the pandemic—i.e., in or after 1918—and zero otherwise. The baseline treatment “Excess Deaths” is defined in equation (3). In columns (1) and (4), the dependent variable is the share of individuals employed in STEM occupations. In columns (2) and (5), the dependent variable is the share of individuals employed in medical occupations. In columns (3) and (6), the dependent variable is the share of individuals employed in a STEM non-medical occupation. In columns (1) to (3), the share is computed with respect to the full population. In columns (4) to (6), we restrict the denominator to include only individuals employed in skilled occupations. All specifications include county fixed effects, state-by-year-fixed effects, and control for an interaction term between the population in 1910 and a post-treatment indicator. Standard errors, clustered at the county level, are reported in parentheses. Counties are weighted by population in 1910. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.11: Impact of the Pandemic on Innovation: Weighted Regressions

	Extensive Margin			Intensive Margin		
	(1)	(2)	(3)	(4)	(5)	(6)
	All Patents	Pharmaceutical	Non-Pharmaceutical	All Patents	Pharmaceutical	Non-Pharmaceutical
Excess Deaths $\times$ Post	0.965*** (0.170)	0.808*** (0.131)	0.869*** (0.165)	1.268*** (0.411)	1.244*** (0.307)	1.149*** (0.403)
County FE	Yes	Yes	Yes	Yes	Yes	Yes
State-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Number of Counties	1,275	1,275	1,275	265	265	265
Observations	38,250	38,250	38,250	7,950	7,950	7,950
R ²	0.957	0.932	0.955	0.952	0.914	0.948
Std. Beta Coef.	0.906	2.094	0.896	0.445	0.924	0.431

Notes: This table displays the impact of exposure to the Great Influenza Pandemic on employment in scientific occupations. The unit of observation is a county, observed at yearly frequency between 1900 and 1929. “Post” is a categorical variable equal to one during and after the pandemic—i.e., in or after 1918—and zero otherwise. The baseline treatment “Excess Deaths” is defined in equation (3). In columns (1) and (4), the dependent variable is the share of individuals employed in STEM occupations. In columns (2) and (5), the dependent variable is the share of individuals employed in medical occupations. In columns (3) and (6), the dependent variable is the share of individuals employed in a STEM non-medical occupation. In columns (1) to (3), the share is computed with respect to the full population. In columns (4) to (6), we restrict the denominator to include only individuals employed in skilled occupations. All specifications include county fixed effects, state-by-year-fixed effects, and control for an interaction term between the population in 1910 and a post-treatment indicator. Standard errors, clustered at the county level, are reported in parentheses. Counties are weighted by population in 1910. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.12: Impact of the Pandemic on Employment in Scientific Occupations: Excluding Internal Migrants

	STEM Employment Share		Medical Employment Share	
	(1)	(2)	(3)	(4)
	Full Sample	High-Skilled	Full Sample	High-Skilled
Excess Deaths $\times$ Post	0.282*** (0.042)	5.226*** (1.402)	0.084*** (0.015)	1.662** (0.673)
County FE	Yes	Yes	Yes	Yes
State-Decade FE	Yes	Yes	Yes	Yes
Number of Counties	1,274	1,274	1,274	1,274
Observations	3,822	3,822	3,822	3,822
Sample	Full	Skilled	Full	Skilled
R ²	0.834	0.725	0.785	0.746
Std. Beta Coef.	0.743	0.860	0.572	0.566

Notes: This table displays the impact of the pandemic on the probability of being employed in a STEM occupation. The observation unit is a county at decade frequency between 1900 and 1930. The dependent variable is the share of individuals employed in STEM occupations relative to the overall population (column 1) or the number of people employed in skilled occupations (column 2), the share of individuals employed in medical occupations relative to the overall population (column 3) or the number of people employed in skilled occupations (column 4). We exclude from the sample internal migrants, defined as those who were born in a different state relative to where they are recorded in the 1930 census. STEM, medical, and skilled occupations listed in Table B.2. The baseline treatment “Excess Deaths” is defined in equation (3) and is interacted with a post-pandemic indicator. All specifications include county and state-by-decade fixed effects and further control for an interaction term between the population in 1910 and a post-treatment indicator. Standard errors are clustered at the county level and are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.13: Impact of the Pandemic on Innovation: Robustness Regressions

	All Patents		f (All Patents)		
	(1)	(2)	(3)	(4)	(5)
			Count	$\text{arcsinh}(\cdot)$	Poisson
Excess Deaths Dummy $\times$ Post	0.080***				
	(0.019)				
Excess Deaths $\times$ Post		0.336***	3.736**	0.410***	1.260***
		(0.056)	(1.790)	(0.069)	(0.226)
WW1 Deaths $\times$ Post		3.326			
		(6.144)			
County FE	Yes	Yes	Yes	Yes	Yes
State-Year FE	Yes	Yes	Yes	Yes	Yes
Number of Counties	1,275	1,275	1,275	1,275	1,256
Observations	38,250	38,250	38,250	38,250	37,680
R ²	0.858	0.858	0.911	0.840	0.907
Std. Beta Coef.	0.027	0.155	0.054	0.159	0.018

Notes: This table displays the impact of exposure to the Influenza on innovation. The unit of observation is a county observed at yearly frequency between 1900 and 1929. In columns (1–2), the dependent variable is $\log(1 + \text{total number of patents})$. “Post” is a categorical variable equal to one during and after the pandemic—i.e., in or after 1918—and zero otherwise. The baseline treatment “Excess Deaths” is defined in equation (3). Column (1) reports the results when the treatment is coded as a binary variable equal to one if the continuous variable is above its median and zero otherwise. Columns (2) controls for WW1 deaths interacted with the post-treatment indicator. The dependent variable in column (3) is the raw patent count. In column (4), the dependent variable is the inverse hyperbolic sine of the raw count of patents. In column (5), the model is estimated as a Poisson regression, and the dependent variable is the raw patent count. All specifications include county and state-by-year-fixed effects and control for an interaction term between the population in 1910 and a post-treatment indicator. Standard errors are clustered at the county level and are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.14: Impact of the Pandemic on Breakthrough Innovation

	(1)	(2)
	Breakthrough	Share Breakthrough
Excess Deaths $\times$ Post	0.166*** (0.043)	0.016** (0.008)
County FE	Yes	Yes
State-Year FE	Yes	Yes
Number of Counties	1,275	1,275
Observations	38,250	38,250
R ²	0.797	0.265
Mean Dep. Var.	0.336	0.052
Std. Beta Coef.	0.125	0.087

Notes: This table displays the impact of the Influenza on patent importance. The unit of observation is a county observed at a yearly frequency between 1900 and 1929. “Post” is a categorical variable equal to one during and after the pandemic—i.e., in or after 1918—and zero otherwise. The baseline treatment “Excess Deaths” is defined in equation (3). Importance measures are from Kelly et al. (2021). They measure the “importance” of a patent based on the textual similarity between that patent and previous and future patents. In column (1), the dependent variable is the $\log(1 + \text{number of breakthrough patents})$, defined as those in the top 20% of the patent importance distribution. In column (2), the dependent variable is the $\log(1 + \text{share of breakthrough patents})$. In both cases, we add one to the denominator to ensure we do not drop county-year pairs with no issued patents. Regressions include county and state-by-year-fixed effects and control for an interaction term between the population in 1910 and a post-treatment indicator. Standard errors are clustered at the county level and displayed in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.15: The Impact of the Pandemic on Religiosity and Innovation

	Pre Pandemic	Post Pandemic	Pooled	DiD
	(1)	(2)	(3)	(4)
Patents per Capita				
Religiosity	-0.059*	0.084**	-0.207***	-0.649***
	(0.035)	(0.041)	(0.042)	(0.241)
Religiosity $\times$ Post			0.659***	
			(0.084)	
Religiosity $\times$ Excess Deaths				0.404*
				(0.211)
Excess Deaths $\times$ Post				0.053***
				(0.014)
Excess Deaths $\times$ Religiosity $\times$ Post				0.556***
				(0.072)
County FE	Yes	Yes	Yes	Yes
State–Year FE	Yes	Yes	Yes	Yes
Number of Counties	1,274	1,274	1,274	1,274
Observations	22,932	15,288	38,220	38,220
R ²	0.548	0.734	0.578	0.580

Notes: This table displays the correlation between innovation and religiosity by exposure to the pandemic. The dependent variable is the number of patents divided by county population in 1910, expressed in 1,000 units, and winsorized at the top and bottom 1% level. The unit of observation is a county observed at a yearly frequency between 1900 and 1929. “Post” is a categorical variable equal to one during and after the pandemic—i.e., in or after 1918—and zero otherwise. The baseline treatment “Excess Deaths” is defined in equation (3). In column (1), we display the correlation between religiosity and innovation before the influenza (before 1918); in column (2), we replicate this exercise for the post-pandemic years; in column (3), we pool the years together, and interact religiosity with a post-pandemic indicator; column (4) reports the differential effect of the pandemic by religiosity. All specifications include county and state-by-year-fixed effects and control for an interaction term between the population in 1910 and a post-treatment indicator. Standard errors are clustered at the county level and are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.16: Impact of the Pandemic on Occupational Choice by Religious Background

	(1)	(2)
	Employed in STEM	Employed in Skilled Occupation
Excess Deaths $\times$ Post $\times$ High Religious Background	-0.701** (0.337)	-0.001 (0.008)
Individual Controls	Yes	Yes
County–Year FE	Yes	Yes
High Religious Background–Year FE	Yes	Yes
High Religious Background–County FE	Yes	Yes
Number of Counties	1,275	1,275
Observations	5,347,751	5,347,751
R ²	0.016	0.048

Notes: This table displays the impact of exposure to the pandemic on occupational choice by individual-level background religiosity. The unit of observation is an adult. In column (1), the dependent variable “Employed in STEM” is an indicator equal to one if an individual is employed in a STEM occupation—as defined in Table B.2—or zero otherwise. In column (2), the dependent variable “Employed in Skilled Occupation” is an indicator equal to one if an individual is employed in a skilled occupation—as defined in as defined in Table B.2—or zero otherwise. “Post” is an indicator equal to one if an individual is aged between 18 and 25 in 1918 or zero otherwise. The baseline treatment “Excess Deaths” is defined in equation (3). “High Background Religiosity” is an indicator variable taking value one if the religiosity score of the name of the head of the household is equal to or above the median, or zero if it is below the median. The table displays the coefficient of the interaction between these terms. All specifications include a vector of individual controls for the household head: race, urban status, and place of birth as well as county-by-year, high religious background-by-year, and high religious background-by-county fixed effects. The sample includes the entire population. Standard errors are clustered at the county level and are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.17: The Polarization of Religiosity

	Name-Based Religiosity
Excess Deaths $\times$ Post $\times$ 1st Quintile of Background Religiosity	-0.045** (0.018)
Excess Deaths $\times$ Post $\times$ 2nd Quintile of Background Religiosity	-0.024* (0.014)
Excess Deaths $\times$ Post $\times$ 4th Quintile of Background Religiosity	0.017 (0.015)
Excess Deaths $\times$ Post $\times$ 5th Quintile of Background Religiosity	0.054*** (0.020)
Number of Counties	1,275
Observations	4,246,352
R ²	0.074

Notes: This table displays the regression estimates reported in Figure 6. The unit of observation is a head of household, observed once for each child born between 1900 and 1929 in the household. The observations are weighted by the inverse of the number of children in the household. “Post” is a categorical variable equal to zero for children born during and after the pandemic—i.e., in or after 1918—or zero for those born before the pandemic—i.e., before 1918. The baseline treatment “Excess Deaths” is defined in equation (3). Background religiosity is measured as the religiosity score of the child’s father. The third quintile of background religiosity serves as the baseline. The dependent variable “Name-Based Religiosity” is defined as the religiosity score associated with the child’s name. The specification includes individual fixed effects at the household-head level: race, urban status, and place of birth. It also includes county-by-cohort, county-by-quintiles of background religiosity, and quintiles of background religiosity-by-cohort fixed effects. Standard errors are clustered at the county level and are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

C Summary of Robustness Analyses

Panel A: Religiosity		
Exhibit	Topic	Description
1. Is our name-based measure indeed capturing religiosity?		
i) Table B.6	Accounting for birth order and fertility	One concern is that our results are driven by: i) firstborns, who may be more likely to be named after grandparents (who may have more religious names); ii) numerous families having idiosyncratic naming patterns correlated with religiosity; iii) more religious families having higher fertility. To address these concerns, in Table B.6, we show that our results hold when we drop firstborns (column 1); we drop children beyond the fourth (column 2); we compute household-level average religiosity by assigning to every child a weight that is inversely related to the number of children in the household (column 3).
ii) Table B.7	Religiosity scores without county FE	Our results may be sensitive to how we compute religiosity scores. In Table B.7, columns (1–2) show that our findings hold when we drop county-fixed effects from the measurement equation. Religiosity scores computed this way reflect the stock of religiosity in a given county rather than its change.
iii) Table B.7	Alternative name frequency thresholds	In our main analysis, we impose a threshold of 0.3% for names to be included in the sample for which we estimate religiosity scores. Our results are robust when using alternative frequency thresholds.
iv) Table B.8	Alternative religiosity measures	In Table B.8, we compute a religiosity indicator of first names based on the names of biblical figures and saints collected by Abramitzky et al. (2016). Our results are robust to using this measure.
v) Figure B.1	Predicted vs. measured religiosity	We find a positive correlation between predicted religiosity and religiosity reported in the Census of Religious Bodies.

vi)	Figure B.4	Saint or biblical names	Our measure of religiosity is highly and positively correlated with the one developed by Abramitzky et al. (2016).
-----	------------	-------------------------	--

2. Was the increase in religiosity caused by WW1?

i)	Table B.5, Figure B.3	WW1 and religiosity	In column (2), we interact WW1 deaths with a post-pandemic indicator. The estimated coefficient is not significant. Additionally, in Figure B.3, we show that WW1 deaths and our measure of excess deaths are not correlated.
----	--------------------------	---------------------	---

3. Was the increase in religiosity driven by migrations?

i)	Table B.5	Internal migrants	We check that the effect of the influenza on religiosity is not driven by individuals who migrated to their residence in 1930 from another state, perhaps responding to a slacker labor market following the pandemic (column 3).
ii)	Table B.5	Religiosity of adults	Our results may be driven by selective migrations of more religious individuals towards areas more affected by the pandemic. In column (4), we use the religiosity score of own-name (a measure of an individual's background religiosity) as the dependent variable. As this variable was determined before the pandemic, a significant effect of the Influenza would suggest internal or international migrations correlated with background religiosity and the intensity of the shock. We do not find evidence in support of this mechanism.

4. Is the observation unit (counties) driving the results?

i)	Table B.9	Religiosity of cities	The baseline analysis is performed at the level of counties. We estimate the effect of the influenza on religiosity at the city level using the sample of Clay et al. (2019). We replicate the main results in column 1.
----	-----------	-----------------------	--

5. Are the results driven by particular states?

i)	Figure B.6a	Exclude one state at a time	We exclude one state at a time from the estimation sample and show that the estimated treatment effect remains stable irrespective of the excluded state.
----	-------------	-----------------------------	---

6. Are our results sensible to violations of parallel trends?

i)	Figure B.7a	Parallel trends violations	We employ the methodology developed by Rambachan and Roth (2023) and find that the estimated treatment effect remains robust when we allow for substantial deviations from the baseline parallel trends assumption.
----	-------------	----------------------------	---

Panel B: Innovation

Exhibit	Topic	Description
7. Is our measure indeed capturing innovation activity?		
i)	Table B.13	Alternative measures of innovation
		In columns (3) to (5), we use alternative transformations of the raw patent count (explained in the header of Tables B.13). The results are quantitatively stable across specifications.
8. Was the increase in innovation caused by WW1?		
i)	Table B.13, Figure B.3	WW1 and innovation
		In column (2), we interact WW1 deaths with a post-pandemic indicator. The estimated coefficient is not significant. Additionally, in Figure B.3, we show that WW1 and influenza-related deaths are uncorrelated.
9. Did the pandemic trigger an increase in high- (or low-) impact innovation?		
i)	Table B.14	Breakthrough innovation
		Using the measure defined by Kelly et al. (2021), we show that the number of “breakthrough” patents, <i>i.e.</i> those in the upper 20% of the patent impact distribution, as well as its share with respect to all patents, increases in counties hit harder by the pandemic.
10. Is the observation unit (counties) driving the results?		

i)	Table B.9	Innovation of cities	The baseline analysis is performed at the county level. We estimate the effect of the influenza on innovation at the city level using the sample of Clay et al. (2019). We replicate the main results in column (2).
----	-----------	----------------------	--

11. Was the increase in STEM driven by migrations?

i)	Table B.12	Excluding internal migrants	One results may be driven by selective migrations of STEM individuals towards areas more affected by the pandemic. We explicitly test this by excluding individuals who were born in a different state than the one where they were recorded in the 1930 census. Results remain under both sample restrictions.
----	------------	-----------------------------	---

12. Are our results driven by particular states?

i)	Figure B.6b	Exclude one state at a time	We exclude one state at a time from the estimation sample and show that the estimated treatment effect remains stable irrespective of the excluded state.
----	-------------	-----------------------------	---

13. Are our results sensible to violations of parallel trends?

i)	Figure B.7b	Parallel trends violations	We employ the methodology developed by Rambachan and Roth (2023) and find that the estimated treatment effect remains robust when we allow for substantial deviations from the baseline parallel trends assumption.
----	-------------	----------------------------	---
